

Active Filter Design Techniques

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20.1 Introduction

What is a filter?

A filter is a device that passes electric signals at certain frequencies or frequency ranges while preventing the passage of others.—Webster.

Filter circuits are used in a wide variety of applications. In the field of telecommunication, bandpass filters are used in the audio frequency range (0 kHz to 20 kHz) for modems and speech processing. High frequency bandpass filters (several hundred megahertz) are used for channel selection in telephone central offices. Data acquisition systems usually require antialiasing low pass filters as well as low pass noise filters in their preceding signal conditioning stages. System power supplies often use band rejection filters to suppress the 60 Hz line frequency and high frequency transients.

In addition, there are filters that do not filter frequencies of a complex input signal but just add a linear phase shift to each frequency component, thus contributing to a constant time delay. These are called *all pass filters*.

At high frequencies (>1 MHz), all these filters usually consist of passive components such as inductors (L), resistors (R), and capacitors (C). They are then called *LRC filters*.

In the lower frequency range (1 Hz to 1 MHz), however, the inductor value becomes very large and the inductor itself gets quite bulky, making economical production difficult.

In these cases, active filters become important. Active filters are circuits that use an operational amplifier (op amp) as the active device in combination with some resistors and capacitors to provide an LRC-like filter performance at low frequencies (Figure 20.1).

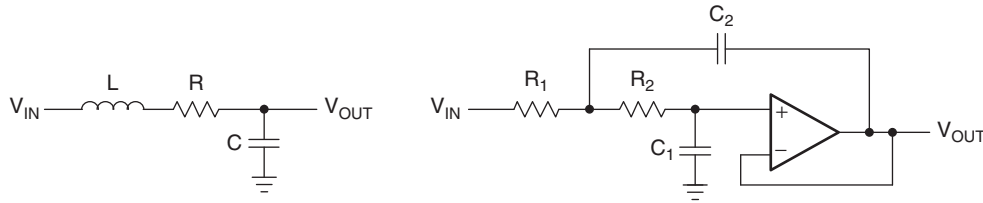


Figure 20.1: Second order passive low pass and second order active low pass filters.

This chapter covers active filters. It introduces the three main filter optimizations (Butterworth, Tschebyscheff, and Bessel), followed by five sections describing the most common active filter applications: low pass, high pass, bandpass, band rejection, and all pass filters. Rather than resembling just another filter book, the individual filter sections are written in a cookbook style, thus avoiding tedious mathematical derivations. Each section starts with the general transfer function of a filter, followed by the design equations to calculate the individual circuit components. The chapter closes with a section on practical design hints for single supply filter designs.

20.2 Fundamentals of Low Pass Filters

The most simple low pass filter is the passive RC low pass network shown in Figure 20.2.

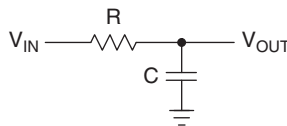


Figure 20.2: First order passive RC low pass filter.

Its transfer function is

$$A(S) = \frac{\frac{1}{RC}}{S + \frac{1}{RC}} = \frac{1}{1 + SRC}$$

where the complex frequency variable, $S = j\omega + \sigma$, allows for any time variable signals. For pure sine waves, the damping constant, σ , becomes zero and $S = j\omega$.

For a normalized presentation of the transfer function, S is referred to the *filter's corner frequency* or *-3 dB frequency*, ω_C , and has these relationships:

$$S = \frac{S}{\omega_C} = \frac{j\omega}{\omega_C} = j\frac{f}{f_C} = j\Omega$$

With the corner frequency of the low pass in [Figure 20.2](#) being $f_C = 1/2\pi RC$, S becomes $S = sRC$ and the transfer function $A(s)$ results in

$$A(S) = \frac{1}{1 + S}$$

The magnitude of the gain response is

$$|A| = \frac{1}{\sqrt{1 + \Omega^2}}$$

For frequencies $\Omega \gg 1$, the roll-off is 20 dB/decade. For a steeper roll-off, n filter stages can be connected in series, as shown in [Figure 20.3](#). To avoid loading effects, op amps, operating as impedance converters, separate the individual filter stages.

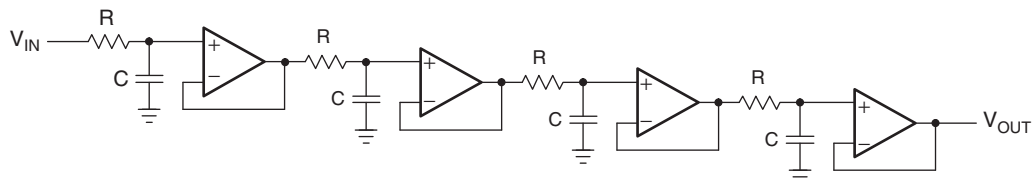


Figure 20.3: Fourth order passive RC low pass filter with decoupling amplifiers.

The resulting transfer function is

$$A(S) = \frac{1}{(1 + \alpha_1 S)(1 + \alpha_2 S) \dots (1 + \alpha_N S)}$$

In the case that all filters have the same cutoff frequency, f_C , the coefficients become $\alpha_1 = \alpha_2 = \dots = \alpha_n = \alpha\sqrt[n]{2 - 1}$, and f_C of each partial filter is $1/\alpha$ times higher than f_C of the overall filter.

Figure 20.4 shows the results of a fourth order RC low pass filter. The roll-off of each partial filter (Curve 1) is -20 dB/decade, increasing the roll-off of the overall filter (Curve 2) to 80 dB/decade.

Note: Filter response graphs plot gain versus the normalized frequency axis $\Omega(\Omega = f/f_C)$.

The corner frequency of the overall filter is reduced by a factor of $\alpha \approx 2.3$ times versus the -3 dB frequency of partial filter stages.

In addition, Figure 20.4 shows the transfer function of an ideal fourth order low pass function (Curve 3).

In comparison to the ideal low pass filter, the RC low pass filter lacks the following characteristics:

- The passband gain varies long before the corner frequency, f_C , thus amplifying the upper passband frequencies less than the lower passband.
- The transition from the passband into the stop band is not sharp but happens gradually, moving the actual 80 dB roll-off by 1.5 octaves above f_C .
- The phase response is not linear, thus increasing the amount of signal distortion significantly.

The gain and phase response of a low pass filter can be optimized to satisfy *one* of the following three criteria:

1. A maximum passband flatness.
2. An immediate passband to stop band transition.
3. A linear phase response.

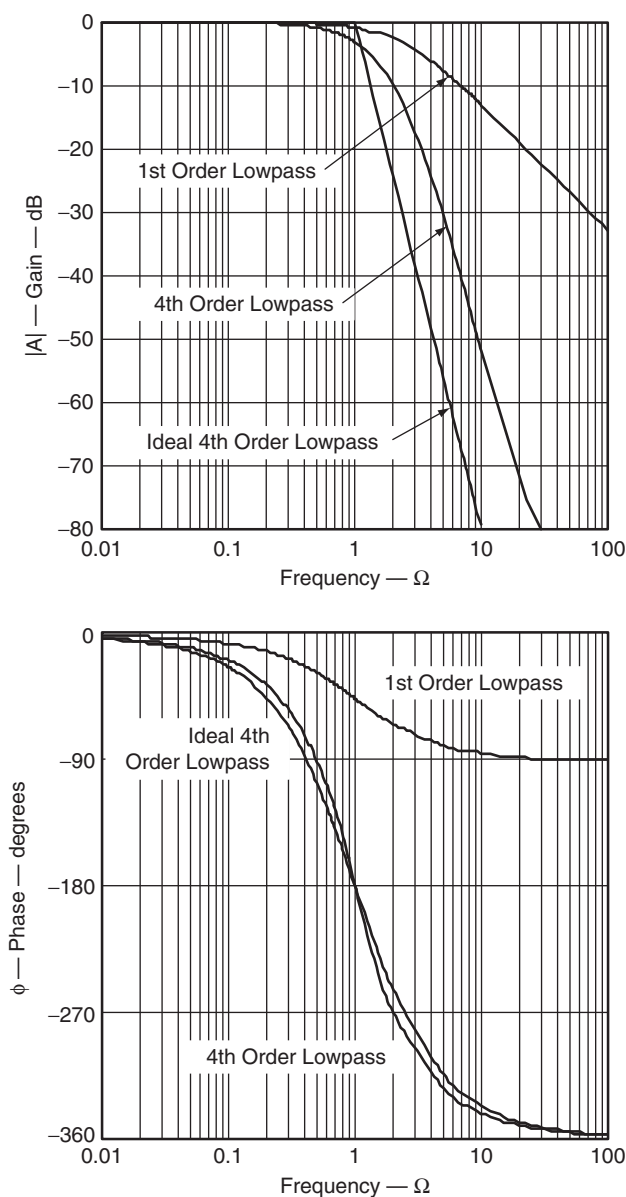


Figure 20.4: Frequency and phase responses of a fourth order passive RC low pass filter. Curve 1, first order partial low pass filter; Curve 2, fourth order overall low pass filter; Curve 3, ideal fourth order low pass filter.

For that purpose, the transfer function must allow for complex poles and needs to be of the following type:

$$A(S) = \frac{A_0}{(1 + a_1S + b_1S^2)(1 + a_2S + b_2S^2) \dots (1 + a_nS + b_nS^2)} = \frac{A_0}{\prod_i (1 + a_iS + b_iS^2)}$$

where A_0 is the passband gain at DC and a_i and b_i are the filter coefficients.

Since the denominator is a product of quadratic terms, the transfer function represents a series of cascaded second order low pass stages, with a_i and b_i being positive real coefficients. These coefficients define the complex pole locations for each second order filter stage, thus determining the behavior of its transfer function.

The following three types of predetermined filter coefficients are available listed in table format in [Section 20.9](#):

- The Butterworth coefficients, optimizing the passband for maximum flatness.
- The Tschhebyscheff coefficients, sharpening the transition from passband into the stop band.
- The Bessel coefficients, linearizing the phase response up to f_C .

The transfer function of a passive RC filter does not allow further optimization, due to the lack of complex poles. The only possibility to produce conjugate complex poles using passive components is the application of LRC filters. However, these filters are mainly used at high frequencies. In the lower frequency range (<10 MHz), the inductor values become very large and the filter becomes uneconomical to manufacture. In these cases, active filters are used. Active filters are RC networks that include an active device, such as an operational amplifier (op amp).

[Section 20.3](#) shows that the products of the RC values and the corner frequency must yield the predetermined filter coefficients a_i and b_i to generate the desired transfer function.

The following subsections introduce the most commonly used filter optimizations.

20.2.1 Butterworth Low Pass Filters

The Butterworth low pass filter provides maximum passband flatness. Therefore, a Butterworth low pass is often used as antialiasing filter in data converter applications where precise signal levels are required across the entire passband.

Figure 20.5 plots the gain response of different orders of Butterworth low pass filters versus the normalized frequency axis, $\Omega(\Omega = f/f_C)$; the higher the filter order, the longer the passband flatness.

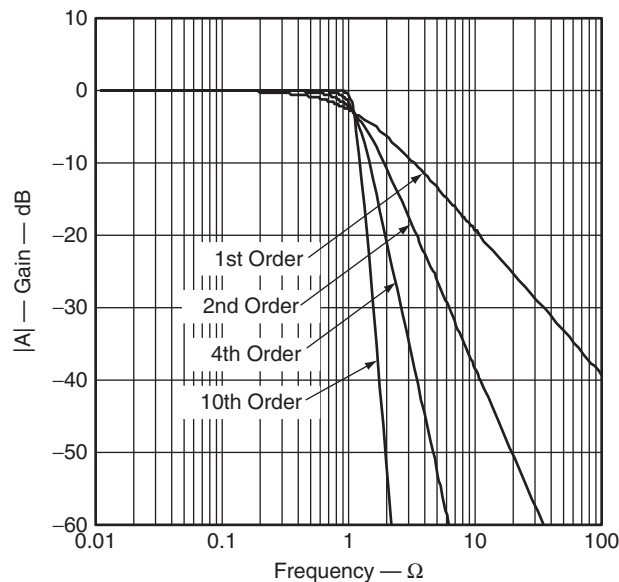


Figure 20.5: Amplitude responses of Butterworth low pass filters.

20.2.2 Tschebyscheff Low Pass Filters

The Tschebyscheff low pass filters provide an even higher gain roll-off above f_C . However, as Figure 20.6 shows, the passband gain is not monotone but contains ripples of constant magnitude instead. For a given filter order, the higher the passband ripples, the higher the filter's roll-off.

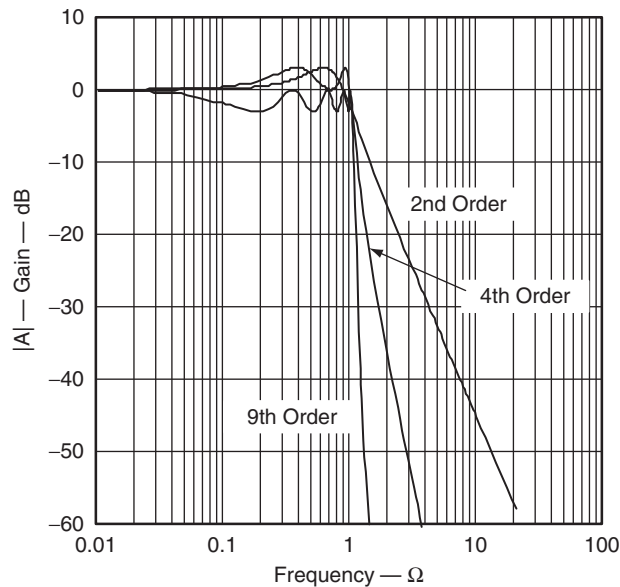


Figure 20.6: Gain responses of Tschebyscheff low pass filters.

With increasing filter order, the influence of the ripple magnitude on the filter roll-off diminishes.

Each ripple accounts for one second order filter stage. Filters with even order numbers generate ripples above the 0 dB line, while filters with odd order numbers create ripples below 0 dB.

Tschebyscheff filters are often used in filter banks, where the frequency content of a signal is of more importance than a constant amplification.

20.2.3 Bessel Low Pass Filters

The Bessel low pass filters have a linear phase response (Figure 20.7) over a wide frequency range, which results in a constant group delay (Figure 20.8) in that frequency range. Bessel low pass filters, therefore, provide an optimum square wave transmission behavior. However, the passband gain of a Bessel low pass filter is not as flat as that of the Butterworth low pass, and the transition from passband to stop band is by far not as sharp as that of a Tschebyscheff low pass filter (Figure 20.9).

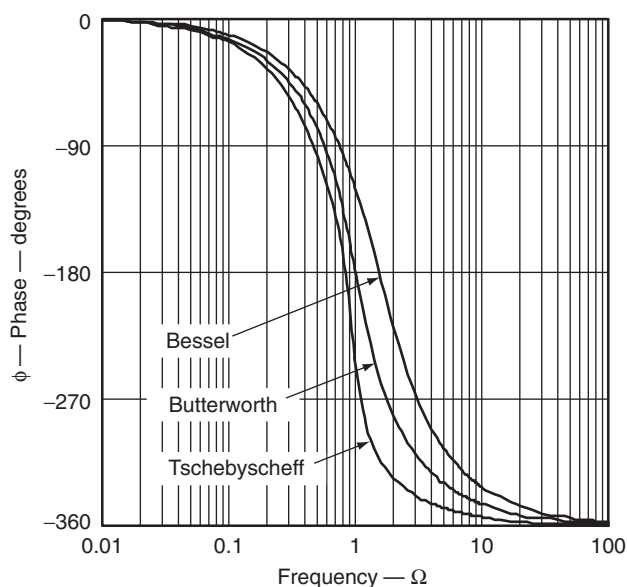


Figure 20.7: Comparison of phase responses of fourth order low pass filters.

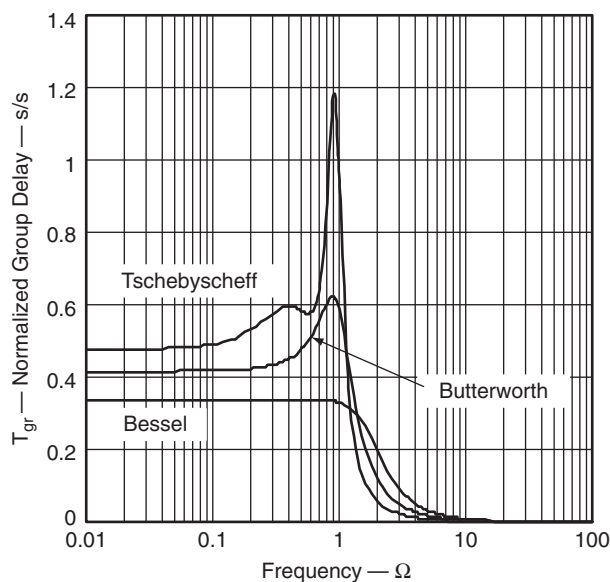


Figure 20.8: Comparison of normalized group delay (T_{gr}) of fourth order low pass filters.

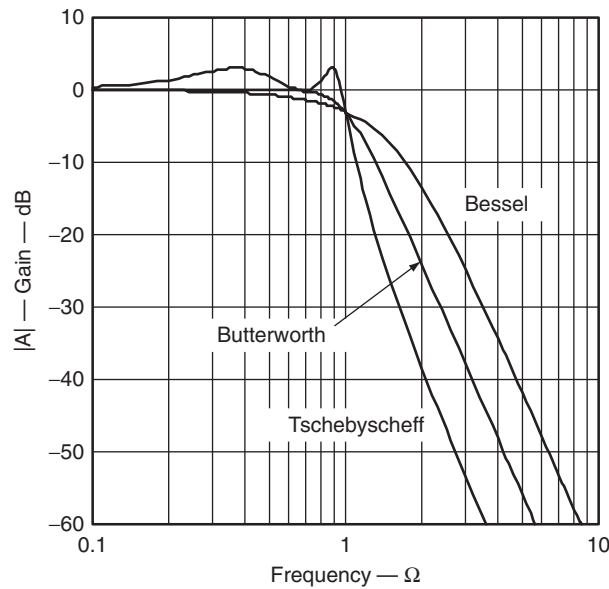


Figure 20.9: Comparison of gain responses of fourth order low pass filters.

20.2.4 Quality Factor Q

The quality factor Q is an equivalent design parameter to the filter order n . Instead of designing an n th order Tschebyscheff low pass filter, the problem can be expressed as designing a Tschebyscheff low pass filter with a certain Q .

For bandpass filters, Q is defined as the ratio of the mid frequency, f_m , to the bandwidth at the two -3 dB points:

$$Q = \frac{f_m}{(f_2 - f_1)}$$

For low pass and high pass filters, Q represents the pole quality and is defined as

$$Q = \frac{\sqrt{b_i}}{a_i}$$

High values of Q can be graphically presented as the distance between the 0 dB line and the peak point of the filter's gain response. An example is given in [Figure 20.10](#), which shows a 10th order Tschebyscheff low pass filter and its five partial filters with their individual Q values.

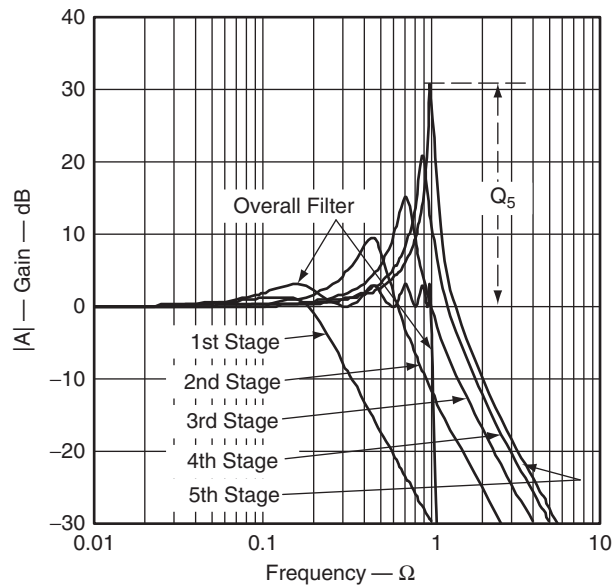


Figure 20.10: Quality factor Q on a 10th order Tschebyscheff low pass filter.

The gain response of the fifth filter stage peaks at 31 dB, which is the logarithmic value of Q_5 :

$$Q_5(\text{dB}) = 20 \log Q_5$$

Solving for the numerical value of Q_5 yields

$$Q_5 = 10^{\frac{31}{20}} = 35.48$$

which is within 1% of the theoretical value of $Q = 35.85$ given in [Section 20.9](#), [Table 20.12](#), last row.

The graphical approximation is good for $Q > 3$. For lower Q values, the graphical values differ from the theoretical value significantly. However, only higher Q values are of concern, since the higher the Q is, the more a filter inclines to instability.

20.2.5 Summary

The general transfer function of a low pass filter is

$$A(S) = \frac{A_0}{\prod_i (1 + a_i S + b_i S^2)} \quad (20.1)$$

The filter coefficients a_i and b_i distinguish between Butterworth, Tschebyscheff, and Bessel filters. The coefficients for all three types of filters are tabulated down to the 10th order in [Section 20.9](#), [Tables 20.7 through 20.13](#).

The multiplication of the denominator terms with each other yields an n th order polynomial of S , with n being the filter order.

While n determines the gain roll-off above f_C with $-n \times 20$ dB/decade, a_i and b_i determine the gain behavior in the passband.

In addition, the ratio $\sqrt{b_i}/a_i = Q$ is defined as the pole quality. The higher the Q value, the more a filter inclines to instability.

20.3 Low Pass Filter Design

[Equation \(20.1\)](#) represents a cascade of second order low pass filters. The transfer function of a single stage is

$$A_i(S) = \frac{A_0}{(1 + a_i S + b_i S^2)} \quad (20.2)$$

For a first order filter, the coefficient b is always zero ($b_1 = 0$), yielding

$$A(S) = \frac{A_0}{1 + a_1 S} \quad (20.3)$$

The first order and second order filter stages are the building blocks for higher order filters.

Often, the filters operate at unity gain ($A_0 = 1$) to lessen the stringent demands on the op amp's open loop gain.

Figure 20.11 shows the cascading of filter stages up to the sixth order. A filter with an even order number consists of second order stages only, while filters with an odd order number include an additional first order stage at the beginning.

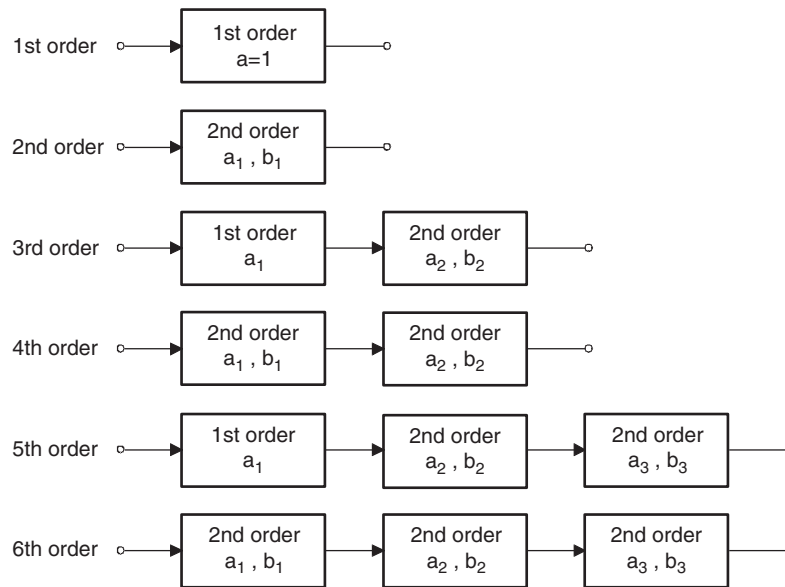


Figure 20.11: Cascading filter stages for higher order filters.

Figure 20.10 demonstrates that the higher the corner frequency of a partial filter, the higher is its Q . Therefore, to avoid the saturation of the individual stages, the filters need to be placed in the order of rising Q values. The Q values for each filter order are listed (in rising order) in Section 20.9, Tables 20.7 through 20.13.

20.3.1 First Order Low Pass Filter

Figures 20.12 and 20.13 show a first order low pass filter in the inverting and noninverting configurations.

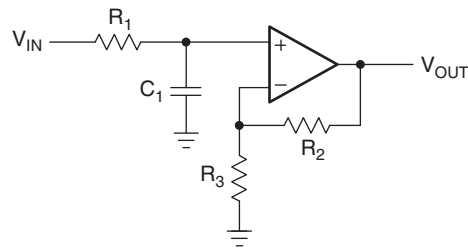


Figure 20.12: First order noninverting low pass filter.

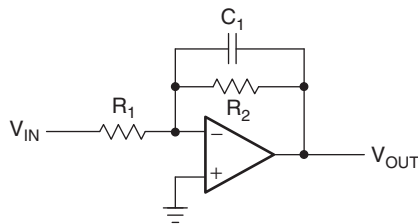


Figure 20.13: First order inverting low pass filter.

The transfer functions of the circuits are

$$A(S) = \frac{1 + \frac{R_2}{R_3}}{1 + \omega_c R_1 C_1 S}$$

and

$$A(S) = \frac{-\frac{R_2}{R_1}}{1 + \omega_c R_2 C_1 S}$$

The negative sign indicates that the inverting amplifier generates a 180° phase shift from the filter input to the output.

The coefficient comparison between the two transfer functions and Equation (20.3) yields

$$A_0 = 1 + \frac{R_2}{R_3} \text{ and } A_0 = -\frac{R_2}{R_1}$$

$$a_1 = \omega_c R_1 C_1 \text{ and } a_1 = \omega_c R_2 C_1$$

To dimension the circuit, specify the corner frequency (f_c), the DC gain (A_0), and capacitor C_1 , then solve for resistors R_1 and R_2 :

$$R_1 = \frac{a_1}{2\pi f_c C_1} \text{ and } R_2 = \frac{a_1}{2\pi f_c C_1}$$

$$R_2 = R_3(A_0 - 1) \text{ and } R_1 = -\frac{R_2}{A_0}$$

The coefficient a_1 is taken from one of the coefficient tables (Tables 20.7 through 20.13 in Section 20.9).

Note that all filter types are identical in their first order and $a_1 = 1$. For higher filter orders, however, $a_1 \neq 1$, because the corner frequency of the first order stage is different from the corner frequency of the overall filter.

Example 20.1. First Order Unity Gain Low Pass Filter For a first order unity gain low pass filter with $f_c = 1$ kHz and $C_1 = 47$ nF, R_1 calculates to

$$R_1 = \frac{a_1}{2\pi f_c C_1} = \frac{1}{2\pi \times 10^3 \text{ Hz} \times 47 \times 10^{-9} \text{ F}} = 3.38 \text{ k}\Omega$$

However, to design the first stage of a third order unity gain Bessel low pass filter, assuming the same values for f_c and C_1 , requires a different value for R_1 . In this case, obtain a_1 for a third order Bessel filter from Table 20.7 in Section 20.9 (Bessel coefficients) to calculate R_1 :

$$R_1 = \frac{a_1}{2\pi f_c C_1} = \frac{0.756}{2\pi \times 10^3 \text{ Hz} \times 47 \times 10^{-9} \text{ F}} = 2.56 \text{ k}\Omega$$

When operating at unity gain, the noninverting amplifier reduces to a voltage follower (Figure 20.14), thus inherently providing a superior gain accuracy. In the case of the inverting amplifier, the accuracy of the unity gain depends on the tolerance of the two resistors, R_1 and R_2 .

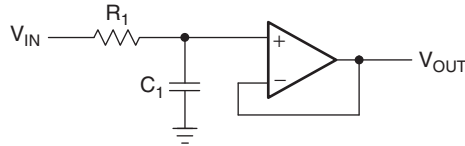


Figure 20.14: First order noninverting low pass filter with unity gain.

20.3.2 Second Order Low Pass Filter

There are two topologies for a second order low pass filter: the Sallen-Key and the multiple feedback (MFB).

Sallen-Key Topology

The general Sallen-Key topology in Figure 20.15 allows for separate gain setting via $A_0 = 1 + R_4/R_3$. However, the unity gain topology in Figure 20.16 is usually applied in filter designs with high gain accuracy, unity gain, and low Q values ($Q < 3$).

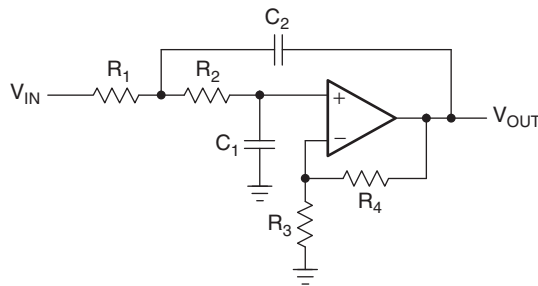


Figure 20.15: General Sallen-Key low pass filter.

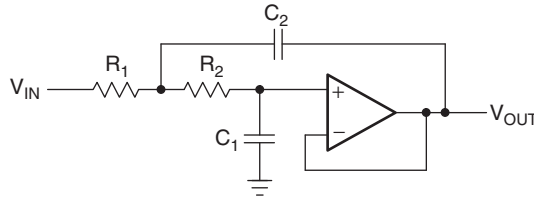


Figure 20.16: Unity gain Sallen-Key low pass filter.

The transfer function of the circuit in [Figure 20.15](#) is

$$A(S) = \frac{A_0}{1 + \omega_c [C_1(R_1 + R_2) + (1 - A_0)R_1C_2]S + \omega_c^2 R_1 R_2 C_1 C_2 S^2}$$

For the unity gain circuit in [Figure 20.16](#) ($A_0 = 1$), the transfer function simplifies to

$$A(S) = \frac{1}{1 + \omega_c C_1 (R_1 + R_2)S + \omega_c^2 R_1 R_2 C_1 C_2 S^2}$$

The coefficient comparison between this transfer function and [Equation \(20.2\)](#) yields

$$A_0 = 1$$

$$a_1 = \omega_c C_1 (R_1 + R_2)$$

$$b_1 = \omega_c^2 R_1 R_2 C_1 C_2$$

Given C_1 and C_2 , the resistor values for R_1 and R_2 are calculated through

$$R_{1,2} = \frac{a_1 C_2 \pm \sqrt{a_1^3 C_2^2 - 4b_1 C_1 C_2}}{4\pi f_c C_1 C_2}$$

To obtain real values under the square root, C_2 must satisfy the following condition:

$$C_2 \geq C_1 \frac{4b_1}{a_1^2}$$

Example 20.2. Second Order Unity Gain Tschebyscheff Low Pass Filter The task is to design a second order unity gain Tschebyscheff low pass filter with a corner frequency of $f_C = 3$ kHz and a 3 dB passband ripple.

From Table 20.12 (the Tschebyscheff coefficients for 3 dB ripple), obtain the coefficients a_1 and b_1 for a second order filter with $a_1 = 1.0650$ and $b_1 = 1.9305$.

Specifying C_1 as 22 nF yields a C_2 value of

$$C_2 \geq C_1 \frac{4b_1}{a_1^2} = 22 \cdot 10^{-9} \text{ nF} \times \frac{4 \times 1.9305}{1.065^2} \cong 150 \text{ nF}$$

Inserting a_1 and b_1 into the resistor equation for $R_{1,2}$ results in

$$R_1 = \frac{1.065 \times 150 \times 10^{-9} - \sqrt{(1.065 \times 150 \times 10^{-9})^2 - 4 \times 1.9305 \times 22 \times 10^{-9} \times 150 \times 10^{-9}}}{4\pi \times 3 \times 10^3 \times 22 \times 10^{-9} \times 150 \times 10^{-9}} = 1.26 \text{ k}\Omega$$

and

$$R_2 = \frac{1.065 \times 150 \times 10^{-9} + \sqrt{(1.065 \times 150 \times 10^{-9})^2 - 4 \times 1.9305 \times 22 \times 10^{-9} \times 150 \times 10^{-9}}}{4\pi \times 3 \times 10^3 \times 22 \times 10^{-9} \times 150 \times 10^{-9}} = 1.30 \text{ k}\Omega$$

with the final circuit shown in Figure 20.17.

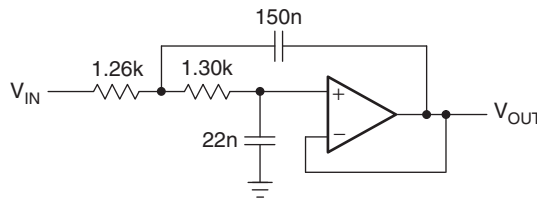


Figure 20.17: Second order unity gain Tschebyscheff low pass with 3 dB ripple.

A special case of the general Sallen-Key topology is the application of equal resistor values and equal capacitor values: $R_1 = R_2 = R$ and $C_1 = C_2 = C$.

The general transfer function changes to

$$A(S) = \frac{A_0}{1 + \omega_c RC(3 - A_0)S + (\omega_c RC)^2 S^2} \text{ with } A_0 = 1 + \frac{R_4}{R_3}$$

The coefficient comparison with Equation (20.2) yields

$$a_1 = \omega RC(3 - A_0)$$

$$b_1 = (\omega RC)^2$$

Given C and solving for R and A_0 results in

$$R = \frac{\sqrt{b_1}}{2\pi f_c C}$$

and

$$A_0 = 3 - \frac{a_1}{\sqrt{b_1}} = 3 - \frac{1}{Q}$$

Therefore, A_0 depends solely on the pole quality Q and vice versa; Q , and with it the filter type, is determined by the gain setting of A_0 :

$$Q = \frac{1}{3 - A_0}$$

The circuit in Figure 20.18 allows the filter type to be changed through the various resistor ratios R_4/R_3 .

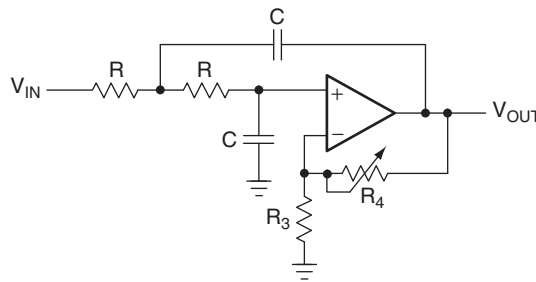


Figure 20.18: Adjustable second order low pass filter.

Table 20.1 lists the coefficients of a second order filter for each filter type and gives the resistor ratios that adjust the Q value.

Table 20.1: Second Order Filter Coefficients

Second order	Bessel	Butterworth	3 dB Tschebyscheff
a_1	1.3617	1.4142	1.065
b_1	0.618	1	1.9305
Q	0.58	0.71	1.3
R_4/R_3	0.268	0.568	0.234

Multiple Feedback Topology

The MFB topology is commonly used in filters that have high Q values and require a high gain. The transfer function of the circuit in Figure 20.19 is

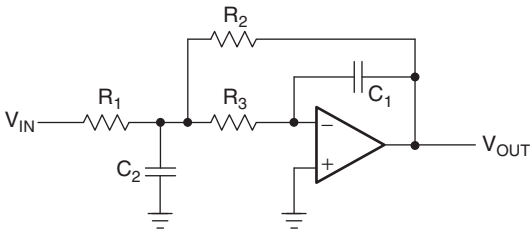


Figure 20.19: Second order MFB low pass filter.

$$A(S) = -\frac{\frac{R_2}{R_1}}{1 + \omega_c C_1 \left(R_2 + R_3 + \frac{R_2 R_3}{R_1} \right) S + \omega_c^2 C_1 C_2 R_2 R_3 S^2}$$

Through coefficient comparison with Equation (20.2), one obtains the relations

$$A_0 = -\frac{R_2}{R_1}$$

$$a_1 = \omega_c C_1 \left(R_2 + R_3 + \frac{R_2 R_3}{R_1} \right)$$

$$b_1 = \omega_c^2 C_1 C_2 R_2 R_3$$

Given C_1 and C_2 and solving for the resistors $R_1 - R_3$,

$$R_2 = \frac{a_1 C_2 - \sqrt{a_1^2 C_2^2 - 4b_1 C_1 C_2 (1 - A_0)}}{4\pi f_c C_1 C_2}$$

$$R_1 = \frac{R_2}{-A_0}$$

$$R_3 = \frac{b_1}{4\pi^2 f_c C_1 C_2 R_2}$$

To obtain real values for R_2 , C_2 must satisfy the following condition:

$$C_2 \geq C_1 \frac{4b_1(1 - A_0)}{a_1^2}$$

20.3.3 Higher Order Low Pass Filter

Higher order low pass filters are required to sharpen a desired filter characteristic. For that purpose, first order and second order filter stages are connected in series, so that the product of the individual frequency responses results in the optimized frequency response of the overall filter.

To simplify the design of the partial filters, the coefficients a_i and b_i for each filter type are listed in the coefficient tables (Tables 20.7 through 20.13 in Section 20.9), with each table providing sets of coefficients for the first 10 filter orders.

Example 20.3. Fifth Order Filter The task is to design a fifth order unity gain Butterworth low pass filter with the corner frequency $f_c = 50$ kHz.

First the coefficients for a fifth order Butterworth filter are obtained from Table 20.8, Section 20.9, and shown here in Table 20.2.

Table 20.2: Example 20.3 Filters

	a_i	b_i
Filter 1	$a_1 = 1$	$b_1 = 0$
Filter 2	$a_2 = 1.6180$	$b_2 = 1$
Filter 3	$a_3 = 0.6180$	$b_3 = 1$

Then, dimension each partial filter by specifying the capacitor values and calculating the required resistor values. The first filter is shown in Figure 20.20.

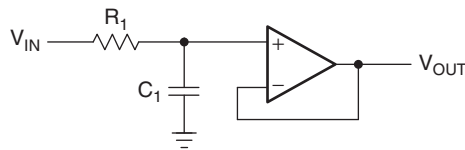


Figure 20.20: First order unity gain low pass filter.

With $C_1 = 1 \text{ nF}$,

$$R_1 = \frac{a_1}{2\pi f_c C_1} = \frac{1}{2\pi \times 50 \times 10^3 \text{ Hz} \times 1 \times 10^{-9} \text{ F}} = 3.18 \text{ k}\Omega$$

The closest 1% value is 3.16 k Ω .

The second filter is shown in Figure 20.21.

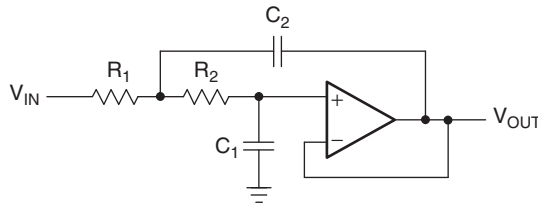


Figure 20.21: Second order unity gain Sallen-Key low pass filter.

With $C_1 = 820 \text{ pF}$,

$$C_2 \geq C_1 \frac{4b_2}{a_2^2} = 820 \times 10^{-12} \text{ F} \times \frac{4 \times 1}{1.618^2} = 1.26 \text{ nF}$$

The closest 5% value is 1.5 nF.

With $C_1 = 820 \text{ pF}$ and $C_2 = 1.5 \text{ nF}$, calculate the values for R_1 and R_2 through

$$R_1 = \frac{a_2 C_2 - \sqrt{a_2^2 C_2^2 - 4b_2 C_1 C_2}}{4\pi f_c C_1 C_2}$$

and

$$R_2 = \frac{a_2 C_2 + \sqrt{a_2^2 C_2^2 - 4b_2 C_1 C_2}}{4\pi f_c C_1 C_2}$$

and obtain

$$R_1 = \frac{1.618 \times 1.5 \times 10^{-9} - \sqrt{(1.618 \times 1.5 \times 10^{-9})^2 - 4 \times 1 \times 820 \times 10^{-12} \times 1.5 \times 10^{-9}}}{4\pi \times 50 \times 10^3 \times 820 \times 10^{-12} \times 1.5 \times 10^{-9}} = 1.87 \text{ k}\Omega$$

$$R_2 = \frac{1.618 \times 1.5 \times 10^{-9} + \sqrt{(1.618 \times 1.5 \times 10^{-9})^2 - 4 \times 1 \times 820 \times 10^{-12} \times 1.5 \times 10^{-9}}}{4\pi \times 50 \times 10^3 \times 820 \times 10^{-12} \times 1.5 \times 10^{-9}} = 4.42 \text{ k}\Omega$$

R_1 and R_2 are available 1% resistors.

The calculation of the third filter is identical to the calculation of the second filter, except that a_2 and b_2 are replaced by a_3 and b_3 , thus resulting in different capacitor and resistor values.

Specify C_1 as 330 pF, and obtain C_2 with

$$C_2 \geq C_1 \frac{4b_3}{a_3^2} = 330 \times 10^{-12} \text{ F} \times \frac{4.1}{0.618^2} = 3.46 \text{ nF}$$

The closest 10% value is 4.7 nF.

With $C_1 = 330$ pF and $C_2 = 4.7$ nF, the values for R_1 and R_2 are

- $R_1 = 1.45$ k Ω , with the closest 1% value being 1.47 k Ω .
- $R_2 = 4.51$ k Ω , with the closest 1% value being 4.53 k Ω .

Figure 20.22 shows the final filter circuit with its partial filter stages.

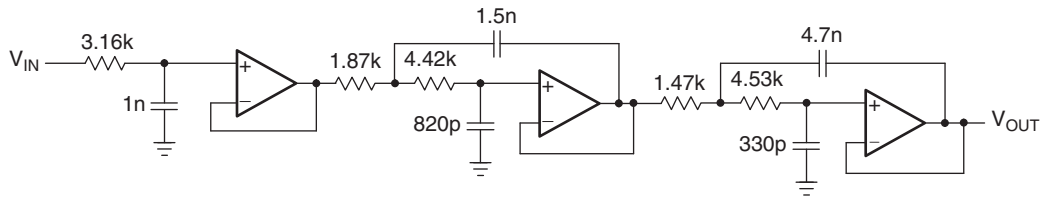


Figure 20.22: Fifth order unity gain Butterworth low pass filter.

20.4 High Pass Filter Design

By replacing the resistors of a low pass filter with capacitors and its capacitors with resistors, a high pass filter is created (Figure 20.23).

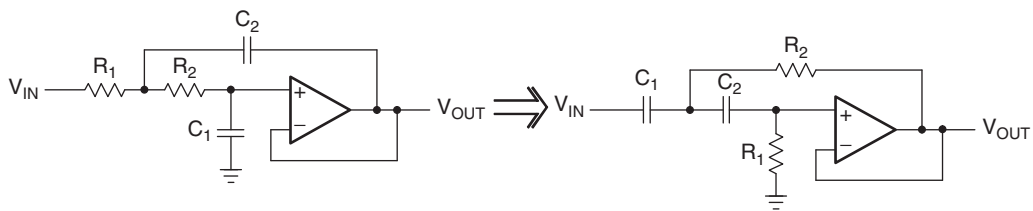


Figure 20.23: Low pass to high pass transition through components exchange.

To plot the gain response of a high pass filter (Figure 20.24), mirror the gain response of a low pass filter at the corner frequency, $\Omega = 1$, thus replacing Ω with $1/\Omega$ and S with $1/S$ in Equation (20.1).

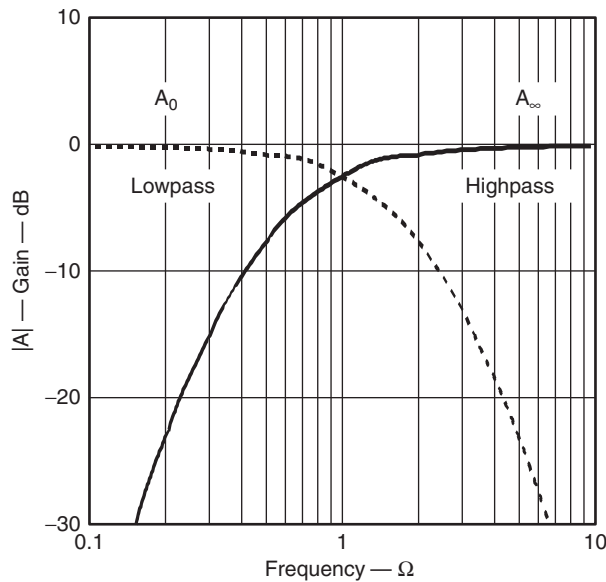


Figure 20.24: Developing the gain response of a high pass filter.

The general transfer function of a high pass filter is then

$$A(S) = \frac{A_{\infty}}{\prod_i \left(1 + \frac{a_i}{S} + \frac{b_i}{S^2} \right)} \quad (20.4)$$

with A_{∞} being the passband gain.

Since Equation (20.4) represents a cascade of second order high pass filters, the transfer function of a single stage is

$$A_i(S) = \frac{A_{\infty}}{\left(1 + \frac{a_i}{S} + \frac{b_i}{S^2} \right)} \quad (20.5)$$

With $b = 0$ for all first order filters, the transfer function of a first order filter simplifies to

$$A(S) = \frac{A_0}{1 + \frac{a_i}{S}} \quad (20.6)$$

20.4.1 First Order High Pass Filter

Figures 20.25 and 20.26 show a first order high pass filter in the noninverting and inverting configurations.

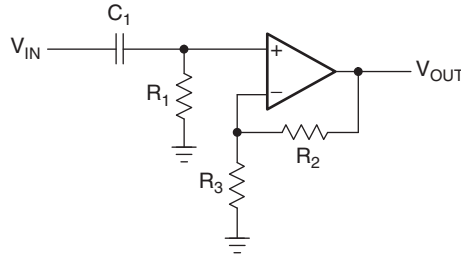


Figure 20.25: First order noninverting high pass filter.

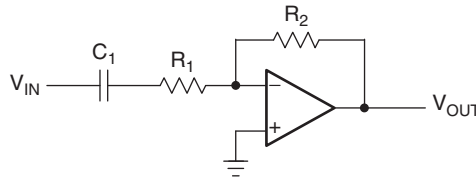


Figure 20.26: First order inverting high pass filter.

The transfer functions of the circuits are

$$A(S) = \frac{1 + \frac{R_2}{R_3}}{1 + \frac{1}{\omega_c R_1 C_1} \times \frac{1}{S}}$$

and

$$A(S) = -\frac{\frac{R_2}{R_1}}{1 + \frac{1}{\omega_c R_1 C_1} \times \frac{1}{S}}$$

The negative sign indicates that the inverting amplifier generates a 180° phase shift from the filter input to the output.

The coefficient comparison between the two transfer functions and Equation (20.6) provides two passband gain factors:

$$A_{\infty} = 1 + \frac{R_2}{R_3}$$

and

$$A_{\infty} = -\frac{R_2}{R_1}$$

while the term for the coefficient a_1 is the same for both circuits:

$$a_1 = \frac{1}{\omega_c R_1 C_1}$$

To dimension the circuit, specify the corner frequency (f_c), the DC gain (A_{∞}), and capacitor (C_1), then solve for R_1 and R_2 :

$$R_1 = \frac{1}{2\pi f_c a_1 C_1}$$

$$R_2 = R_3(A_{\infty} - 1) \text{ and } R_2 = -R_1 A_{\infty}$$

20.4.2 Second Order High Pass Filter

High pass filters use the same two topologies as the low pass filters: Sallen-Key and multiple feedback. The only difference is that the positions of the resistors and the capacitors have changed.

Sallen-Key Topology

The general Sallen-Key topology in Figure 20.27 allows for separate gain setting via $A_0 = 1 + R_4/R_3$.

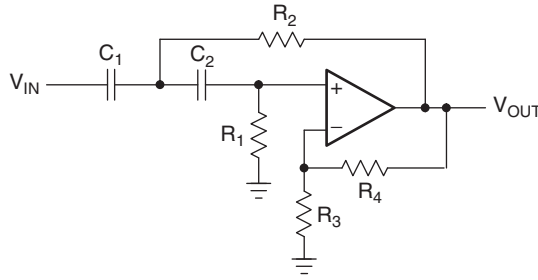


Figure 20.27: General Sallen-Key high pass filter.

The transfer function of the circuit in [Figure 20.27](#) is

$$A(S) = \frac{\alpha}{1 + \frac{R_2(C_1 + C_2) + R_1C_2(1 - \alpha)}{\omega_c R_1 R_2 C_1 C_2} \times \frac{1}{S} + \frac{1}{\omega_c R_1 R_2 C_1 C_2} \times \frac{1}{S^2}}$$

with

$$\alpha = 1 + \frac{R_4}{R_3}$$

The unity gain topology in [Figure 20.28](#) is usually applied in low Q filters with high gain accuracy.

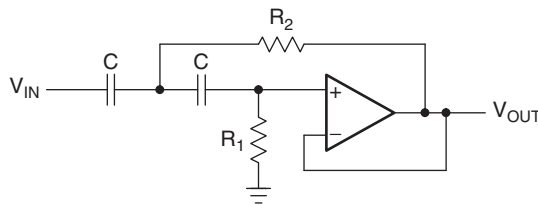


Figure 20.28: Unity gain Sallen-Key high pass filter.

To simplify the circuit design, it is common to choose unity gain ($\alpha = 1$) and $C_1 = C_2 = C$. The transfer function of the circuit in [Figure 20.28](#) then simplifies to

$$A(S) = \frac{1}{1 + \frac{2}{\omega_c R_1 C} \times \frac{1}{S} + \frac{2}{\omega_c^2 R_1 R_2 C^2} \times \frac{1}{S^2}}$$

The coefficient comparison between this transfer function and Equation (20.5) yields

$$A_\infty = 1$$

$$a_1 = \frac{2}{\omega_c R_1 C}$$

$$b_1 = \frac{1}{\omega_c^2 R_1 R_2 C^2}$$

Given C , the resistor values for R_1 and R_2 are calculated through

$$R_1 = \frac{1}{\pi f_c C a_1}$$

$$R_2 = \frac{a_1}{4\pi f_c C b_1}$$

Multiple Feedback Topology

The MFB topology is commonly used in filters that have high Q values and require a high gain.

To simplify the computation of the circuit, capacitors C_1 and C_3 assume the same value ($C_1 = C_3 = C$), as shown in Figure 20.29.

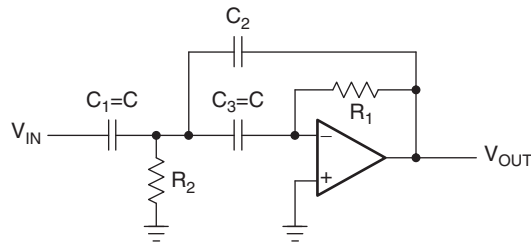


Figure 20.29: Second order MFB high pass filter.

The transfer function of the circuit in [Figure 20.29](#) is

$$A(S) = \frac{-\frac{C}{C_2}}{1 + \frac{2C_2 + C}{\omega_c R_1 C_2 C} \times \frac{1}{S} + \frac{2C_2 + C}{\omega_c^2 R_2 R_1 C_2 C} \times \frac{1}{S^2}}$$

Through coefficient comparison with [Equation \(20.5\)](#), obtain the following relations:

$$\begin{aligned} A_\infty &= \frac{C}{C_2} \\ a_1 &= \frac{2C + C_2}{\omega_c R_1 C C_2} \\ b_1 &= \frac{2C + C_2}{\omega_c R_1 C C_2} \end{aligned}$$

Given capacitors C and C_2 and solving for resistors R_1 and R_2 ,

$$\begin{aligned} R_1 &= \frac{1 - 2A_\infty}{2\pi f_c \times C \times a_1} \\ R_2 &= \frac{a_1}{2\pi f_c \times b_1 C_2 (1 - 2A_\infty)} \end{aligned}$$

The passband gain (A_∞) of an MFB high pass filter can vary significantly due to the wide tolerances of the two capacitors, C and C_2 . To keep the gain variation at a minimum, it is necessary to use capacitors with tight tolerance values.

20.4.3 Higher Order High Pass Filter

As with the low pass filters, higher order high pass filters are designed by cascading first order and second order filter stages. The filter coefficients are the same ones used for the low pass filter design and are listed in the coefficient tables ([Tables 20.7 through 20.13](#) in [Section 20.9](#)).

Example 20.4. Third Order High Pass Filter with $f_C = 1$ kHz The task is to design a third order unity gain Bessel high pass filter with the corner frequency $f_C = 1$ kHz.

Obtain the coefficients for a third order Bessel filter from Table 20.7, Section 20.9, shown here in Table 20.3, and compute each partial filter by specifying the capacitor values and calculating the required resistor values.

Table 20.3: Example 20.4 Filters

	a_i	b_i
Filter 1	$a_1 = 0.756$	$b_1 = 0$
Filter 2	$a_2 = 0.9996$	$b_2 = 0.4772$

For the first filter, with $C_1 = 100$ nF,

$$R_1 = \frac{1}{2\pi f_c a_1 C_1} = \frac{1}{2\pi \times 10^3 \text{ Hz} \times 0.756 \times 100 \times 10^{-9} \text{ F}} = 2.105 \text{ k}\Omega$$

The closest 1% value is 2.1 k Ω .

For the second filter, with $C = 100$ nF,

$$R_1 = \frac{1}{\pi f_c C a_1} = \frac{1}{\pi \times 10^3 \times 100 \times 10^{-9} \times 0.756} = 3.18 \text{ k}\Omega$$

The closest 1% value is 3.16 k Ω .

$$R_2 = \frac{a_1}{4\pi f_c C b_1} = \frac{0.9996}{4\pi \times 10^3 \times 100 \times 10^{-9} \times 0.4772} = 1.67 \text{ k}\Omega$$

The closest 1% value is 1.65 k Ω .

Figure 20.30 shows the final filter circuit.

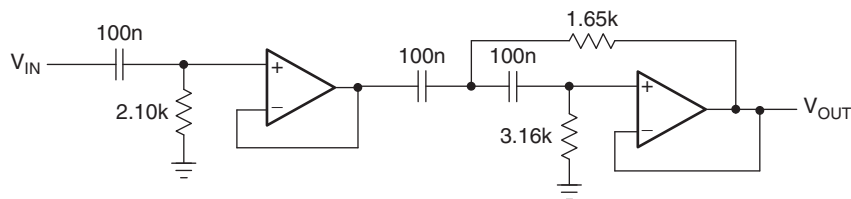


Figure 20.30: Third order unity gain Bessel high pass filter.

20.5 Bandpass Filter Design

In [Section 20.4](#), a high pass response was generated by replacing the term S in the low pass transfer function with the transformation $1/S$. Likewise, a bandpass characteristic is generated by replacing the S term with the transformation

$$\frac{1}{\Delta\Omega} \left(S + \frac{1}{S} \right) \quad (20.7)$$

In this case, the passband characteristic of a low pass filter is transformed into the upper passband half of a bandpass filter. The upper passband is then mirrored at the mid frequency, f_m ($\Omega = 1$), into the lower passband half ([Figure 20.31](#)).

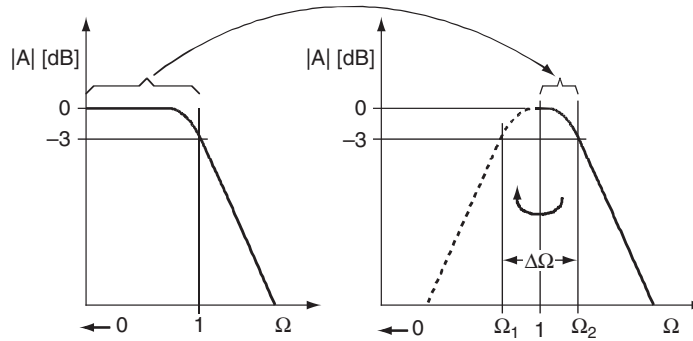


Figure 20.31: Low pass to bandpass transition.

The corner frequency of the low pass filter transforms to the lower and upper -3 dB frequencies of the bandpass, Ω_1 and Ω_2 . The difference between both frequencies is defined as the normalized bandwidth $\Delta\Omega$:

$$\Delta\Omega = \Omega_2 - \Omega_1$$

The normalized mid frequency, where $Q = 1$, is

$$\Omega_m = 1 = \Omega_2 \times \Omega_1$$

In analogy to the resonant circuits, the quality factor Q is defined as the ratio of the mid frequency (f_m) to the bandwidth (B):

$$Q = \frac{f_m}{B} = \frac{f_m}{f_2 - f_1} = \frac{1}{\Omega_2 - \Omega_1} = \frac{1}{\Delta\Omega} \quad (20.8)$$

The simplest design of a bandpass filter is the connection of a high pass filter and a low pass filter in series, which is commonly done in wideband filter applications. Thus, a first order high pass filter and a first order low pass provide a second order bandpass, while a second order high pass filter and a second order low pass result in a fourth order bandpass response.

In comparison to wideband filters, narrowband filters of higher order consist of cascaded second order bandpass filters that use the Sallen-Key or the multiple feedback topology.

20.5.1 Second Order Bandpass Filter

To develop the frequency response of a second order bandpass filter, apply the transformation in [Equation \(20.7\)](#) to a first order low pass transfer function:

$$A(S) = \frac{A_0}{1 + S}$$

Replacing S with

$$\frac{1}{\Delta\Omega} \left(S + \frac{1}{S} \right)$$

yields the general transfer function for a second order bandpass filter:

$$A(S) = \frac{A_0 \times \Delta\Omega \times S}{1 + \Delta\Omega S + S^2} \quad (20.9)$$

When designing bandpass filters, the parameters of interest are the gain at the mid frequency (A_m) and the quality factor (Q), which represents the selectivity of a bandpass filter.

Therefore, replace A_0 with A_m and $\Delta\Omega$ with $1/Q$, Equation (20.7), and obtain

$$A(S) = \frac{\frac{A_m}{Q} \times S}{1 + \frac{1}{Q} \times S + S^2} \quad (20.10)$$

Figure 20.32 shows the normalized gain response of a second order bandpass filter for different Q values.

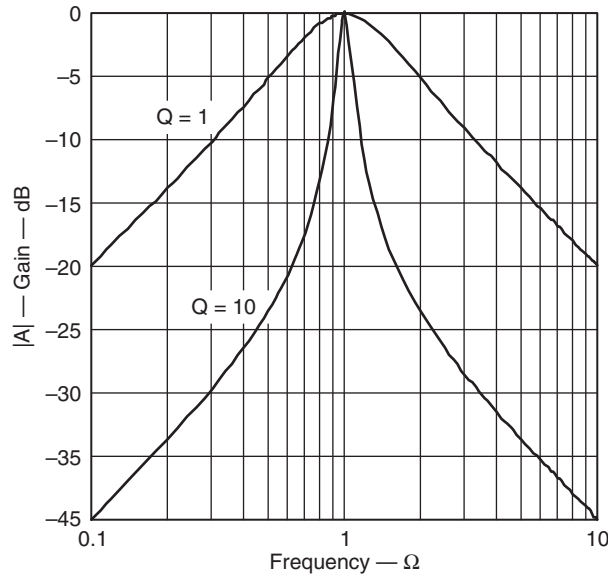


Figure 20.32: Gain response of a second order bandpass filter.

The graph shows that the frequency response of second order bandpass filters gets steeper with rising Q , making the filter more selective.

Sallen-Key Topology

The Sallen-Key bandpass circuit in Figure 20.33 has the following transfer function:

$$A(S) = \frac{G \times RC\omega_m \times S}{1 + RC\omega_m(3 - G) \times S + R^2C^2\omega_m^2 \times S^2}$$

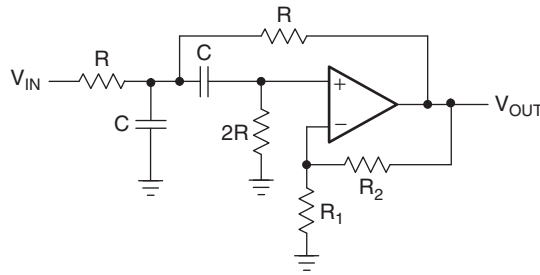


Figure 20.33: Sallen-Key bandpass filter.

Through coefficient comparison with [Equation \(20.10\)](#), obtain the following equations:

$$\text{Mid frequency: } f_m = \frac{1}{2\pi RC}$$

$$\text{Inner gain: } G = 1 + \frac{R_2}{R_1}$$

$$\text{Gain at } f_m: A_m = \frac{G}{3 - G}$$

$$\text{Filter quality: } Q = \frac{1}{3 - G}$$

The Sallen-Key circuit has the advantage that the quality factor (Q) can be varied via the inner gain (G) without modifying the mid frequency (f_m). A drawback is, however, that Q and A_m cannot be adjusted independently.

Care must be taken when G approaches the value of 3, because then A_m becomes infinite and causes the circuit to oscillate.

To set the mid frequency of the bandpass, specify f_m and C then solve for R :

$$R = \frac{1}{2\pi f_m C}$$

Because of the dependency between Q and A_m , there are two options to solve for R_2 , either set the gain at mid frequency,

$$R_2 = \frac{2A_m - 1}{1 + A_m}$$

or design for a specified Q ,

$$R_2 = \frac{2Q - 1}{Q}$$

Multiple Feedback Topology

The MFB bandpass circuit in Figure 20.34 has the following transfer function:

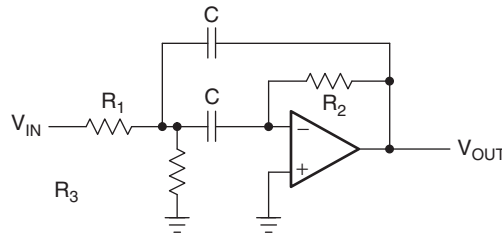


Figure 20.34: MFB bandpass filter.

$$A(S) = \frac{-\frac{R_2 R_3}{R_1 + R_3} C \omega_m \times S}{1 + \frac{2R_1 R_3}{R_1 + R_3} C \omega \times S + \frac{R_1 R_2 R_3}{R_1 + R_3} C^2 \times \omega_m^2 \times S^2}$$

The coefficient comparison with Equation (20.9) yields the following equations:

$$\text{Mid frequency: } f_m = \frac{1}{1\pi C} \sqrt{\frac{R_1 + R_3}{R_1 R_2 R_3}}$$

$$\text{Gain at } f_m: -A_m = \frac{R_2}{2R_1}$$

$$\text{Filter quality: } Q = \pi f_m R_2 C$$

$$\text{Bandwidth: } B = \frac{1}{\pi R_2 C}$$

The MFB bandpass allows adjusting Q , A_m , and f_m independently. The bandwidth and gain factor do not depend on R_3 . Therefore, R_3 can be used to modify the mid frequency without affecting bandwidth, B , or gain, A_m . For low values of Q , the filter can work without R_3 , however, Q then depends on A_m via

$$-A_m = 2Q^2$$

Example 20.5. Second Order MFB Bandpass Filter with $f_m = 1 \text{ kHz}$ To design a second order MFB bandpass filter with a mid frequency of $f_m = 1 \text{ kHz}$, a quality factor of $Q = 10$, and a gain of $A_m = -2$, assume a capacitor value of $C = 100 \text{ nF}$ and solve the previous equations for R_1 through R_3 in the following sequence:

$$R_2 = \frac{Q}{\pi f_m C} = \frac{10}{\pi \times 1 \text{ kHz} \times 100 \text{ nF}} = 31.8 \text{ k}\Omega$$

$$R_1 = \frac{R_2}{-2A_m} = \frac{31.8 \text{ k}\Omega}{4} = 7.96 \text{ k}\Omega$$

$$R_3 = \frac{-A_m R_1}{2Q^2 + A_m} = \frac{2 \times 7.96 \text{ k}\Omega}{200 - 2} = 80.4 \text{ }\Omega$$

20.5.2 Fourth Order Bandpass Filter (Staggered Tuning)

Figure 20.32 shows that the frequency response of second order bandpass filters gets steeper with rising Q . However, some bandpass applications require a flat gain response close to the mid frequency as well as a sharp passband to stop band transition. These tasks can be accomplished by higher order bandpass filters.

Of particular interest is the application of the low pass to bandpass transformation onto a second order low pass filter, since it leads to a fourth order bandpass filter.

Replacing the S term in Equation (20.2) with Equation (20.7) gives the general transfer function of a fourth order bandpass:

$$A(S) = \frac{\frac{S^2 \times A_0(\Delta\Omega)^2}{b_1}}{1 + \frac{a_1}{b_1}\Delta\Omega \times S + \left[2 + \frac{(\Delta\Omega)^2}{b_1}\right] \times S^2 + \frac{a_1}{b_1}\Delta\Omega \times S^3 + S^4} \quad (20.11)$$

Similar to the low pass filters, the fourth order transfer function is split into two second order bandpass terms. Further mathematical modifications yield

$$A(S) = \frac{\frac{A_{mi}}{Q_i} \times \alpha S}{\left[1 + \frac{\alpha S}{Q_1} + (\alpha S)^2\right]} \times \frac{\frac{A_{mi}}{Q_i} \times \frac{S}{\alpha}}{\left[1 + \frac{1}{Q_i} \left(\frac{S}{\alpha}\right) + \left(\frac{S}{\alpha}\right)^2\right]} \quad (20.12)$$

Equation (20.12) represents the connection of two second order bandpass filters in series, where

- A_{mi} is the gain at the mid frequency, f_{mi} , of each partial filter.
- Q_i is the pole quality of each filter.
- α and $1/\alpha$ are the factors by which the mid frequencies of the individual filters, f_{m1} and f_{m2} , derive from the mid frequency, f_m , of the overall bandpass.

In a fourth order bandpass filter with high Q , the mid frequencies of the two partial filters differ only slightly from the overall mid frequency. This method is called *staggered tuning*.

Factor α needs to be determined through successive approximation using Equation (20.13):

$$\alpha^2 = \left[\frac{\alpha \times \Delta\Omega \times a_1}{b_1(1 + \alpha)^2} \right]^2 + \frac{1}{\alpha^2} - 2 - \frac{(\Delta\Omega)^2}{b_1} = 0 \quad (20.13)$$

with a_1 and b_1 being the second order low pass coefficients of the desired filter type.

To simplify the filter design, Table 20.4 lists those coefficients and provides the α values for three quality factors, $Q = 1$, $Q = 10$, and $Q = 100$.

Table 20.4: Values of α for Different Filter Types

Bessel				Butterworth				Tschebyscheff			
a_1	1.3617			a_1	1.4142			a_1	1.0650		
b_1	0.6180			b_1	1.0000			b_1	1.9305		
Q	100	10	1	Q	100	10	1	Q	100	10	1
$\Delta\Omega$	0.01	0.1	1	$\Delta\Omega$	0.01	0.1	1	$\Delta\Omega$	0.01	0.1	1
α	1.0032	1.0324	1.438	α	1.0035	1.036	1.4426	α	1.0033	1.0338	1.39

After α has been determined, all quantities of the partial filters can be calculated using the following equations.

The mid frequency of filter 1 is

$$f_{m1} = \frac{f_m}{\alpha} \quad (20.14)$$

and the mid frequency of filter 2 is

$$f_{m2} = f_m \times \alpha \quad (20.15)$$

with f_m being the mid frequency of the overall forth order bandpass filter.

The individual pole quality, Q_i , is the same for both filters:

$$Q_i = Q \times \frac{(1 + \alpha^2)b_1}{\alpha \times a_1} \quad (20.16)$$

with Q being the quality factor of the overall filter.

The individual gain (A_{mi}) at the partial mid frequencies, f_{m1} and f_{m2} , is the same for both filters:

$$A_{mi} = \frac{Q_i}{Q} \times \sqrt{\frac{A_m}{B_1}} \quad (20.17)$$

with A_m being the gain at mid frequency, f_m , of the overall filter.

Example 20.6. Fourth Order Butterworth Bandpass Filter The task is to design a fourth order Butterworth bandpass filter with the following parameters:

- Mid frequency, $f_m = 10$ kHz.
- Bandwidth, $B = 1000$ Hz.
- Gain, $A_m = 1$.

From [Table 20.4](#), the following values are obtained:

- $a_1 = 1.4142$.
- $b_1 = 1$.
- $\alpha = 1.036$.

In accordance with [Equations \(20.14\) and \(20.15\)](#), the mid frequencies for the partial filters are

$$f_{mi} = \frac{10 \text{ kHz}}{1.036} = 9.653 \text{ kHz}$$

and

$$f_{m2} = 10 \text{ kHz} \times 1.036 = 10.36 \text{ kHz}$$

The overall Q is defined as $Q = f_m/B$, and for this example results in $Q = 10$.

Using [Equation \(20.16\)](#), the Q_i of both filters is

$$Q_i = 10 \times \frac{(1 + 1.036^2) \times 1}{1.036 \times 1.4142} = 14.15$$

With [Equation \(20.17\)](#), the passband gain of the partial filters at f_{m1} and f_{m2} calculates to

$$A_{mi} = \frac{14.15}{10} \times \sqrt{\frac{1}{1}} = 1.415$$

[Equations \(20.16\) and \(20.17\)](#) show that Q_i and A_{mi} of the partial filters need to be independently adjusted. The only circuit that accomplishes this task is the MFB bandpass filter in [Section 20.5.1](#).

To design the individual second order bandpass filters, specify $C = 10 \text{ nF}$, and insert the previously determined quantities for the partial filters into the resistor equations of the MFB bandpass filter. The resistor values for both partial filters are calculated in [Table 20.5](#).

Table 20.5: Resistor Values for Both Partial Filters

Filter 1	Filter 2
$R_{21} = \frac{Q_i}{\pi f_{m1} C} = \frac{14.15}{\pi \times 9.653 \text{ kHz} \times 10 \text{ nF}} = 46.7 \text{ k}\Omega$	$R_{22} = \frac{Q_i}{\pi f_{m2} C} = \frac{14.15}{\pi \times 10.36 \text{ kHz} \times 10 \text{ nF}} = 43.5 \text{ k}\Omega$
$R_{11} = \frac{R_{21}}{-2A_{mi}} = \frac{46.7 \text{ k}\Omega}{-2 \times -1.415} = 16.5 \text{ k}\Omega$	$R_{12} = \frac{R_{22}}{-2A_{mi}} = \frac{43.5}{-2 \times -1.415} = 15.4 \text{ k}\Omega$
$R_{31} = 2 \frac{-A_{mi} R_{11}}{2Q_i^2 + A_{mi}} = \frac{1.415 \times 16.5 \text{ k}\Omega}{2 \times 14.15^2 + 1.415} = 58.1 \text{ }\Omega$	$R_{32} = \frac{A_{mi} R_{12}}{2Q_i^2 + A_{mi}} = \frac{1.415 \times 15.4 \text{ k}\Omega}{2 \times 14.15^2 + 1.415} = 54.2 \text{ }\Omega$

Figure 20.35 compares the gain response of a fourth order Butterworth bandpass filter with $Q = 1$ and its partial filters to the fourth order gain of Example 20.4 with $Q = 10$.

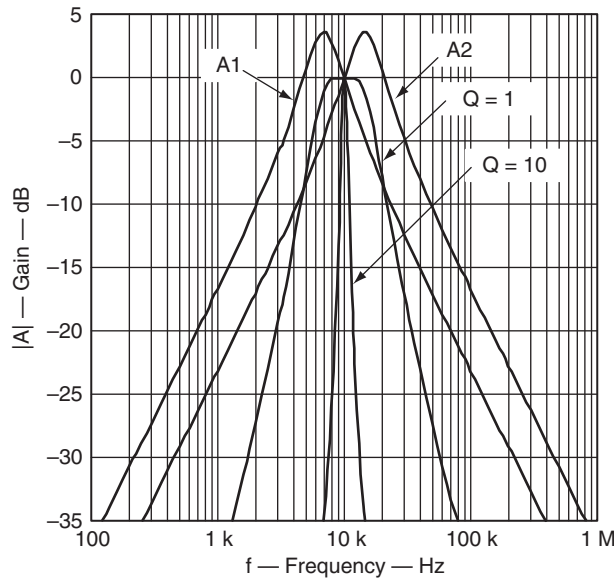


Figure 20.35: Gain responses of a fourth order Butterworth bandpass and its partial filters.

20.6 Band Rejection Filter Design

A band rejection filter is used to suppress a certain frequency rather than a range of frequencies.

Two of the most popular band rejection filters are the active twin T and the active Wien-Robinson circuits, both of which are second order filters.

To generate the transfer function of a second order band rejection filter, replace the S term of a first order low pass response with the transformation in Equation (20.18):

$$\frac{\Delta\Omega}{S + \frac{1}{S}} \quad (20.18)$$

which gives

$$A(S) = \frac{A_0(1 + S^2)}{1 + \Delta\Omega \times S + S^2} \quad (20.19)$$

Thus, the passband characteristic of the low pass filter is transformed into the lower passband of the band rejection filter. The lower passband is then mirrored at the mid frequency, f_m ($\Omega = 1$), into the upper passband half (Figure 20.36).

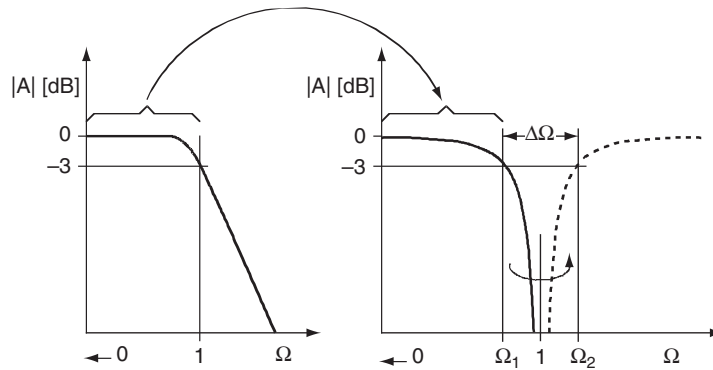


Figure 20.36: Low pass to band rejection transition.

The corner frequency of the low pass filter transforms to the lower and upper -3 dB frequencies of the band rejection filter, Ω_1 and Ω_2 . The difference between both frequencies is the normalized bandwidth $\Delta\Omega$:

$$\Delta\Omega = \Omega_{\max} - \Omega_{\min}$$

Identical to the selectivity of a bandpass filter, the quality of the filter rejection is defined as

$$Q = \frac{f_m}{B} = \frac{1}{\Delta\Omega}$$

Therefore, replacing $\Delta\Omega$ in Equation (20.19) with $1/Q$ yields

$$A(S) = \frac{A_0(1 + S^2)}{1 + \frac{1}{Q} \times S + S^2} \quad (20.20)$$

20.6.1 Active Twin T Filter

The original twin T filter, shown in Figure 20.37, is a passive RC network with a quality factor of $Q = 0.25$. To increase Q , the passive filter is implemented into the feedback loop of an amplifier, turning into an active band rejection filter, shown in Figure 20.38.

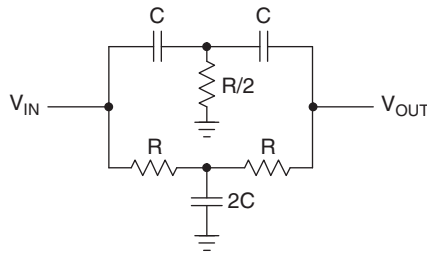


Figure 20.37: Passive twin T filter.

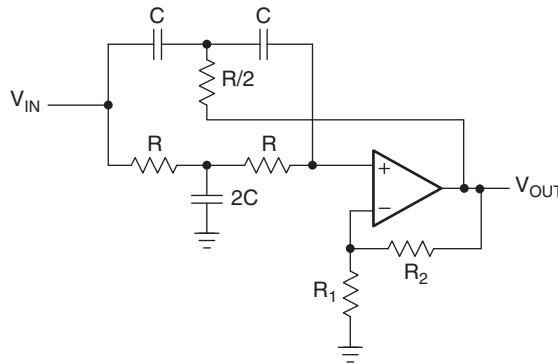


Figure 20.38: Active twin T filter.

The transfer function of the active twin T filter is

$$A(S) = \frac{k(1 + S^2)}{1 + 2(2 - k) \times S + S^2} \quad (20.21)$$

Comparing the variables of Equation (20.21) with Equation (20.20) provides the equations that determine the filter parameters:

$$\text{Mid frequency: } f_m = \frac{1}{2\pi RC}$$

$$\text{Inner gain: } G = 1 + \frac{R_2}{R_1}$$

$$\text{Passband gain: } A_0 = G$$

$$\text{Rejection quality: } Q = \frac{1}{2(2 - G)}$$

The twin T circuit has the advantage that the quality factor (Q) can be varied via the inner gain (G) without modifying the mid frequency (f_m). However, Q and A_m cannot be adjusted independently.

To set the mid frequency of the bandpass, specify f_m and C , then solve for R :

$$R = \frac{1}{2\pi f_m C}$$

Because of the dependency between Q and A_m , there are two options to solve for R_2 , either set the gain at mid frequency,

$$R_2 = (A_0 - 1)R_1$$

or design for a specific Q ,

$$R_2 = R_1 \left(1 - \frac{1}{2Q} \right)$$

20.6.2 Active Wien-Robinson Filter

The Wien-Robinson bridge in Figure 20.39 is a passive band rejection filter with differential output. The output voltage is the difference between the potential of a constant voltage divider and the output of a bandpass filter. Its Q factor is close to that of the twin T circuit. To achieve higher values of Q , the filter is connected into the feedback loop of an amplifier.

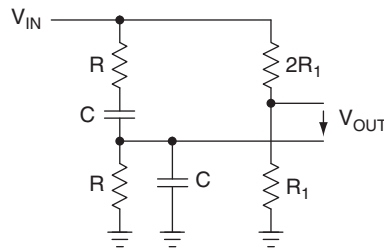


Figure 20.39: Passive Wien-Robinson bridge.

The active Wien-Robinson filter in Figure 20.40 has the transfer function

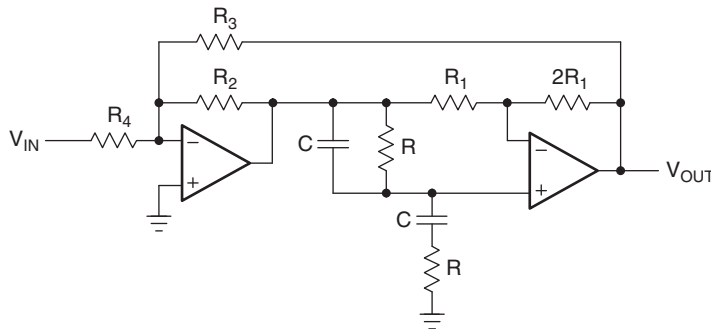


Figure 20.40: Active Wien-Robinson filter.

$$A(S) = -\frac{\frac{\beta}{1+\alpha}(1+S^2)}{1+\frac{3}{1+\alpha} \times S+S^2} \quad (20.22)$$

with $\alpha = R_2/R_3$ and $\beta = R_2/R_4$.

Comparing the variables of Equation (20.22) with Equation (20.20) provides the equations that determine the filter parameters:

$$\text{Mid frequency: } f_m = \frac{1}{2\pi RC}$$

$$\text{Passband gain: } A_0 = -\frac{\beta}{1 + \alpha}$$

$$\text{Rejection quality: } Q = \frac{1 + \alpha}{3}$$

To calculate the individual component values, establish the following design procedure:

1. Define f_m and C and calculate R with

$$R = \frac{1}{2\pi f_m C}$$

2. Specify Q and determine α via $\alpha = 3Q - 1$.
3. Specify A_0 and determine β via $\beta = -A_0 \times 3Q$.
4. Define R_2 and calculate R_3 and R_4 with $R_3 = R_2/\alpha$ and $R_4 = R_2/\beta$.

In comparison to the twin T circuit, the Wien-Robinson filter allows modification of the passband gain, A_0 , without affecting the quality factor, Q .

If f_m is not completely suppressed due to component tolerances of R and C , a fine-tuning of the resistor $2R_2$ is required.

Figure 20.41 shows a comparison between the filter response of a passive band rejection filter with $Q = 0.25$ and an active second order filter with $Q = 1$ and $Q = 10$.

20.7 All Pass Filter Design

In comparison to the previously discussed filters, an all pass filter has a constant gain across the entire frequency range and a phase response that changes linearly with frequency. Because of these properties, all pass filters are used in phase compensation and signal delay circuits.

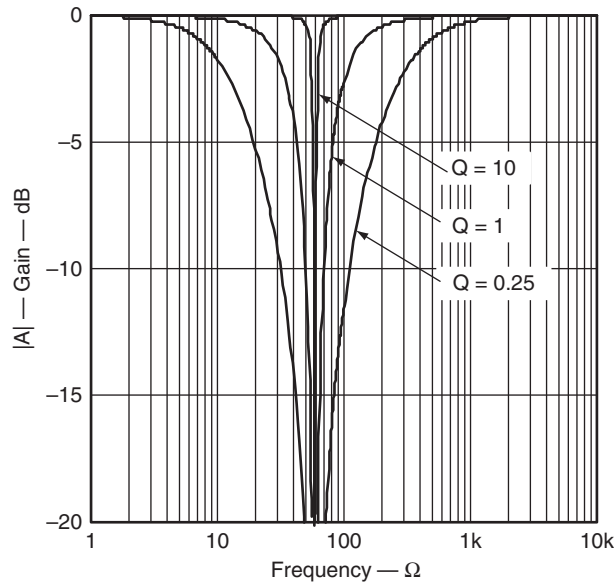


Figure 20.41: Comparison of Q between passive and active band rejection filters.

Similar to the low pass filters, all pass circuits of higher order consist of cascaded first order and second order all pass stages. To develop the all pass transfer function from a low pass response, replace A_0 with the conjugate complex denominator.

The general transfer function of an all pass filter is then

$$A(S) = \frac{\prod_i (1 - a_i S + b_i S^2)}{\prod_i (1 + a_i S + b_i S^2)} \quad (20.23)$$

with a_i and b_i being the coefficients of a partial filter. The all pass coefficients are listed in [Table 20.13](#) of [Section 20.9](#).

Expressing [Equation \(20.23\)](#) in magnitude and phase yields

$$A(s) = \frac{\prod_i \sqrt{(1 - b_i \Omega^2)^2 + a_i^2 \Omega^2} \times e^{-ja}}{\prod_i \sqrt{(1 - b_i \Omega^2)^2 + a_i^2 \Omega^2} \times e^{+ja}} \quad (20.24)$$

This gives a constant gain of 1 and a phase shift, ϕ , of

$$\phi = -2\alpha = -2 \sum_i \arctan \frac{a_i \Omega}{1 - b_i \Omega^2} \quad (20.25)$$

To transmit a signal with minimum phase distortion, the all pass filter must have a constant group delay across the specified frequency band. The group delay is the time by which the all pass filter delays each frequency within that band.

The frequency at which the group delay drops to $1/\sqrt{2}$ times its initial value is the corner frequency, f_c .

The group delay is defined through

$$t_{\text{gr}} = - \frac{d\phi}{d\omega} \quad (20.26)$$

To present the group delay in normalized form, refer t_{gr} to the period of the corner frequency, T_c , of the all pass circuit:

$$T_{\text{gr}} = \frac{t_{\text{gr}}}{T_c} = t_{\text{gr}} \times f_c = t_{\text{gr}} \times \frac{\omega_c}{2\pi} \quad (20.27)$$

Substituting t_{gr} through Equation (20.26) gives

$$T_{\text{gr}} = - \frac{1}{2\pi} \times \frac{d\phi}{d\Omega} \quad (20.28)$$

Inserting the ϕ term in Equation (20.25) into Equation (20.28) and completing the derivation results in

$$T_{\text{gr}} = \frac{1}{\pi} \sum_i \frac{a_i(1 + b_i \Omega^2)}{1 + (a_1^2 - 2b_1) \times \Omega^2 + b_1^2 \Omega^4} \quad (20.29)$$

Setting $\Omega = 0$ in Equation (20.29) gives the group delay for the low frequencies, $0 < \Omega < 1$, which is

$$T_{\text{gr}0} = \frac{1}{\pi} \sum_i a_i \quad (20.30)$$

The values for $T_{\text{gr}0}$ are listed in Table 20.13, Section 20.9, from the first to the tenth order.

In addition, Figure 20.42 shows the group delay response versus the frequency for the first 10 orders of all pass filters.

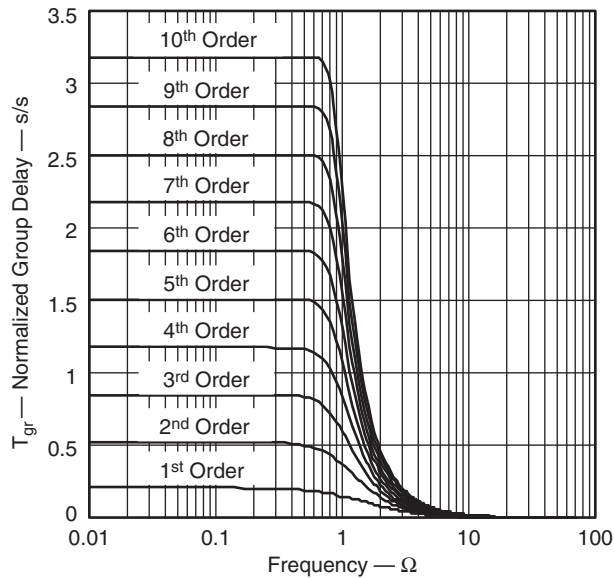


Figure 20.42: Frequency response of the group delay for the first 10 filter orders.

20.7.1 First Order All Pass Filter

Figure 20.43 shows a first order all pass filter with a gain of +1 at low frequencies and a gain of -1 at high frequencies. Therefore, the magnitude of the gain is 1, while the phase changes from 0° to -180° .

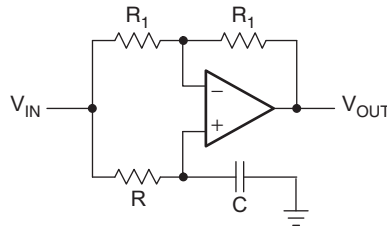


Figure 20.43: First order all pass filter.

The transfer function of this circuit is

$$A(s) = \frac{1 - RC\omega_c \times s}{1 + RC\omega_c \times s}$$

The coefficient comparison with [Equation \(20.23\)](#) ($b_1 = 1$) results in

$$a_i = RC \times 2\pi f_c \quad (20.31)$$

To design a first order all pass, specify f_c and C and then solve for R :

$$R = \frac{a_i}{2\pi f_c \times C} \quad (20.32)$$

Inserting [Equation \(20.31\)](#) into (20.30) and substituting ω_c with [Equation \(20.27\)](#) provides the maximum group delay of a first order all pass filter:

$$t_{gr0} = 2RC \quad (20.33)$$

20.7.2 Second Order All Pass Filter

[Figure 20.44](#) shows that one possible design for a second order all pass filter is to subtract the output voltage of a second order bandpass filter from its input voltage.

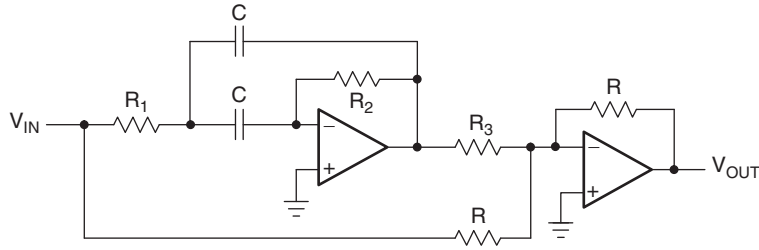


Figure 20.44: Second order all pass filter.

The transfer function of the circuit in [Figure 20.44](#) is

$$A(S) = \frac{1 + (2R_1 - \alpha R_2)C\omega_c \times S + R_1 R_2 C^2 \omega_c^2 \times S^2}{1 + 2R_1 C\omega_c \times S + R_1 R_2 C^2 \omega_c^2 \times S^2}$$

The coefficient comparison with [Equation \(20.23\)](#) yields

$$a_1 = 4\pi f_c R_1 C \quad (20.34)$$

$$b_1 = a_1 \pi f_c R_2 C \quad (20.35)$$

$$\alpha = \frac{a_1^2}{b_1} = \frac{R}{R_3} \quad (20.36)$$

To design the circuit, specify f_c , C , and R , then solve for the resistor values:

$$R_1 = \frac{a_1}{4\pi f_c C} \quad (20.37)$$

$$R_2 = \frac{b_1}{a_1 \pi f_c C} \quad (20.38)$$

$$R_3 = \frac{R}{\alpha} \quad (20.39)$$

Inserting [Equation \(20.34\)](#) into [Equation \(20.30\)](#) and substituting ω_c with [Equation \(20.27\)](#) gives the maximum group delay of a second order all pass filter:

$$t_{gr0} = 4R_1 C \quad (20.40)$$

20.7.3 Higher Order All Pass Filter

Higher order all pass filters consist of cascaded first order and second order filter stages.

Example 20.7. All Pass Filter with a 2 ms Delay A signal with the frequency spectrum $0 < f < 1$ kHz needs to be delayed by 2 ms. To keep the phase distortions at a minimum, the corner frequency of the all pass filter must be $f_c \geq 1$ kHz.

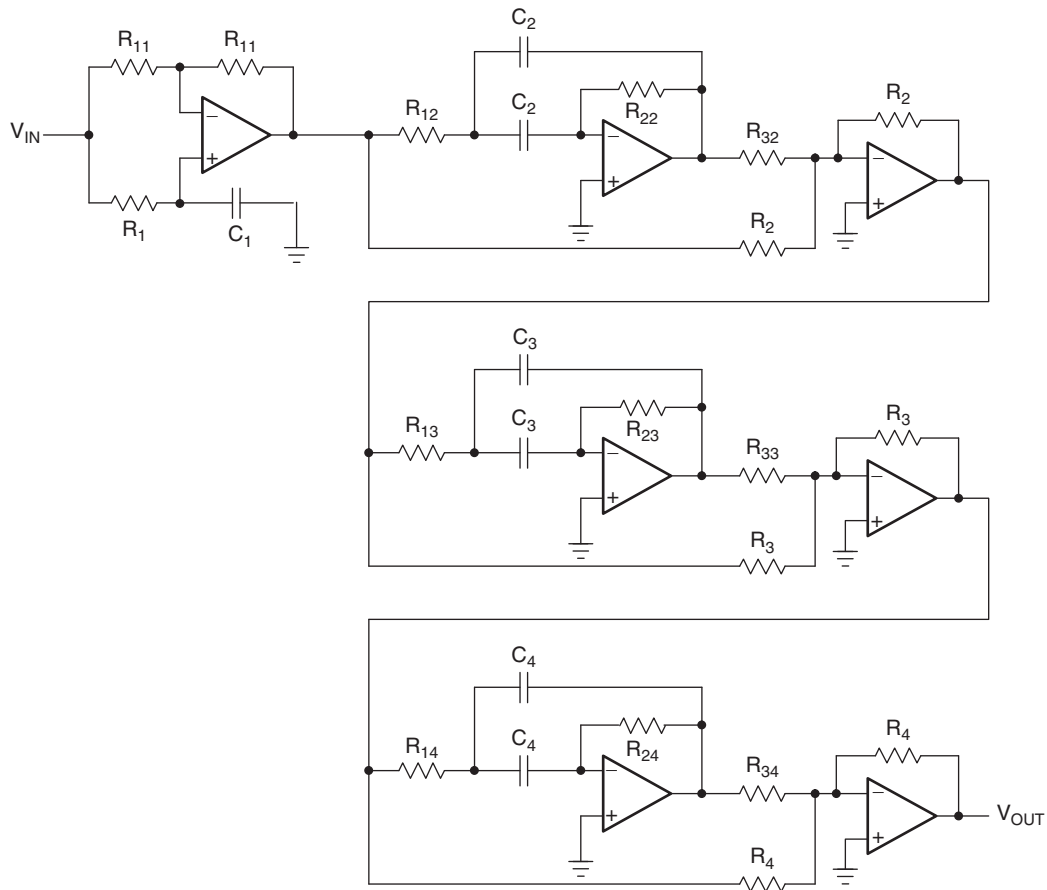


Figure 20.45: Seventh order all pass filter.

Equation (20.27) determines the normalized group delay for frequencies below 1 kHz:

$$T_{\text{gr}0} = \frac{t_{\text{gr}0}}{T_c} = 2 \text{ ms} \times 1 \text{ kHz} = 2.0$$

Figure 20.42 confirms that a seventh order all pass is needed to accomplish the desired delay. The exact value, however, is $T_{\text{gr}0} = 2.1737$. To set the group delay to precisely 2 ms, solve Equation (20.27) for f_C and obtain the corner frequency:

$$f_C = \frac{T_{\text{gr}0}}{t_{\text{gr}0}} = 1.087 \text{ kHz}$$

To complete the design, look up the filter coefficients for a seventh order all pass filter, specify C , and calculate the resistor values for each partial filter.

Cascading the first order all pass with the three second order stages results in the desired seventh order all pass filter.

20.8 Practical Design Hints

This section introduces DC biasing techniques for filter designs in single supply applications, which are usually not required when operating with dual supplies. It also provides recommendations on selecting the type and value range of capacitors and resistors as well as the decision criteria for choosing the correct op amp.

20.8.1 Filter Circuit Biasing

The filter diagrams in this chapter are drawn for dual supply applications. The op amp operates from a positive and a negative supply, while the input and the output voltage are referenced to ground (Figure 20.46).

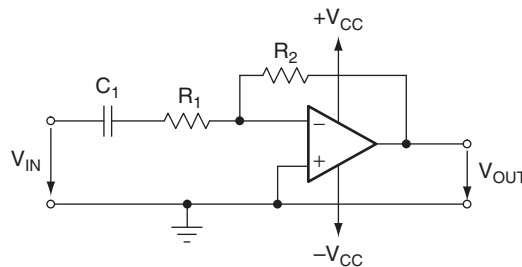


Figure 20.46: Dual supply filter circuit.

For the single supply circuit in Figure 20.47, the lowest supply voltage is ground. For a symmetrical output signal, the potential of the noninverting input is level shifted to midrail.

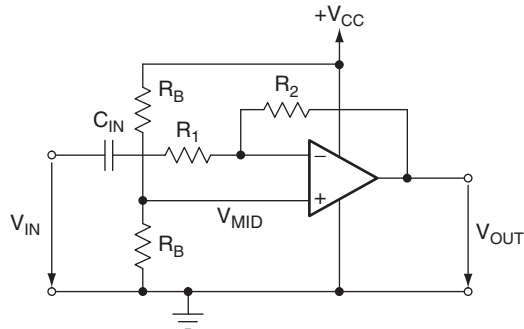


Figure 20.47: Single supply filter circuit.

The coupling capacitor, C_{IN} in Figure 20.47, AC couples the filter, blocking any unknown DC level in the signal source. The voltage divider, consisting of the two equal bias resistors R_B , divides the supply voltage to V_{MID} and applies it to the inverting op amp input.

For simple filter input structures, passive RC networks often provide a low cost biasing solution. In the case of more complex input structures, such as the input of a second order low pass filter, the RC network can affect the filter characteristic. Then, it is necessary to either include the biasing network into the filter calculations or insert an input buffer between biasing network and the actual filter circuit, as shown in Figure 20.48.

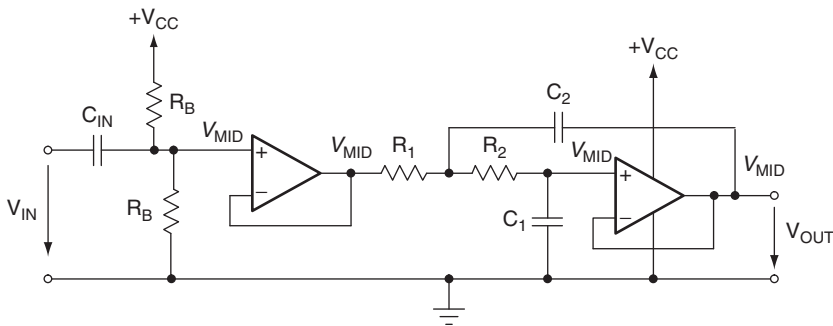


Figure 20.48: Biasing a Sallen-Key low pass filter.

C_{IN} AC-couples the filter, blocking any DC level in the signal source. V_{MID} is derived from V_{CC} via the voltage divider. The op amp operates as a voltage follower and as an impedance converter. V_{MID} is applied via the DC path, R_1 and R_2 , to the noninverting input of the filter amplifier.

Note that the parallel circuit of the resistors, R_B , together with C_{IN} create a high pass filter. To avoid any effect on the low pass characteristic, the corner frequency of the input high pass filter must be low versus the corner frequency of the actual low pass filter.

The use of an input buffer causes no loading effects on the low pass filter, thus keeping the filter calculation simple.

In the case of a higher order filter, all following filter stages receive their bias level from the preceding filter amplifier.

Figure 20.49 shows the biasing of a multiple feedback low pass filter.

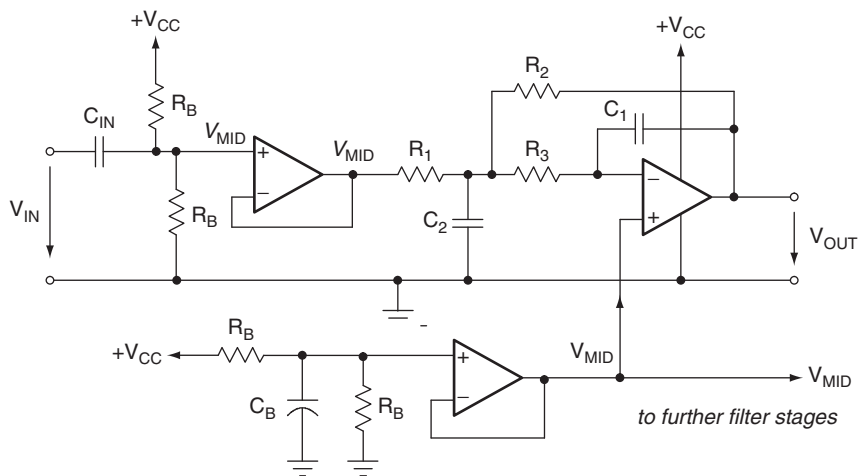


Figure 20.49: Biasing a second order MFB low pass filter.

The input buffer decouples the filter from the signal source. The filter itself is biased via the noninverting amplifier input. For that purpose, the bias voltage is taken from the output of a V_{MID} generator with low output impedance. The op amp operates as a difference amplifier and subtracts the bias voltage of the input buffer from the bias voltage of the V_{MID} generator, thus yielding a DC potential of V_{MID} at zero input signal.

A low cost alternative is to remove the op amp and use a passive biasing network instead. However, to keep loading effects at a minimum, the values for R_B must be significantly higher than without the op amp.

The biasing of a Sallen-Key and an MFB high pass filter is shown in Figure 20.50.

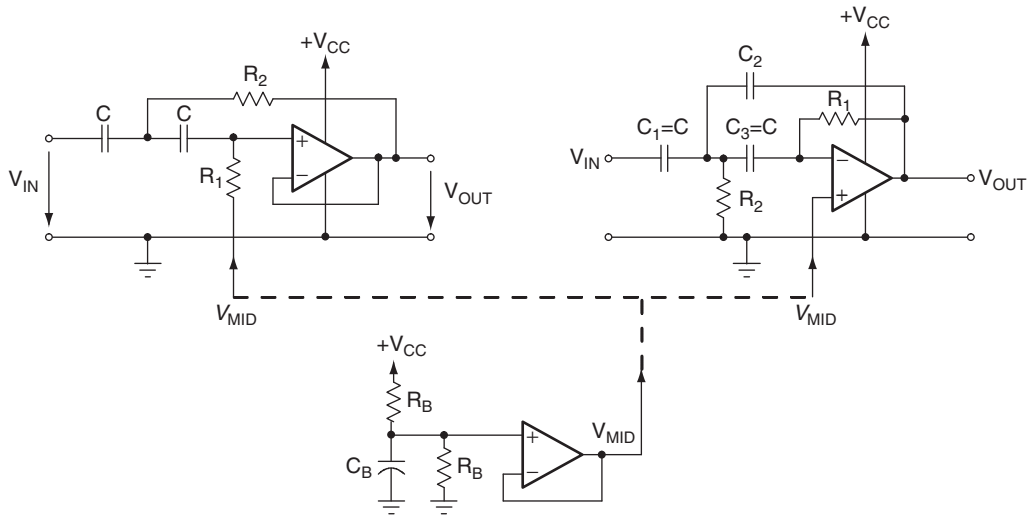


Figure 20.50: Biasing a Sallen-Key and an MFB high pass filter.

The input capacitors of high pass filters already provide the AC coupling between filter and signal source. Both circuits use the V_{MID} generator from Figure 20.50 for biasing. While the MFB circuit is biased at the noninverting amplifier input, the Sallen-Key high pass filter is biased via the only DC path available, which is R_1 . In the AC circuit, the input signals travel via the low output impedance of the op amp to ground.

20.8.2 Capacitor Selection

The tolerance of the selected capacitors and resistors depends on the filter sensitivity and performance. Sensitivity is the measure of the vulnerability of a filter's performance to changes in component values. The important filter parameters to consider are the corner frequency, f_c , and Q .

For example, when Q changes by $\pm 2\%$ due to a $\pm 5\%$ change in the capacitance value, the sensitivity of Q to capacity changes is expressed as $s(Q/C) = 2\%/5\% = 0.4(\%/ \%)$. The following sensitivity approximations apply to second order Sallen-Key and MFB filters:

$$s\frac{Q}{C} \approx s\frac{Q}{C} \approx s\frac{f_c}{C} \approx \pm 0.5 \frac{\%}{\%}$$

Although $0.5\%/ \%$ is a small difference from the ideal parameter, in the case of higher order filters, the combination of small Q and f_c differences in each partial filter can significantly modify the overall filter response from its intended characteristic.

Figures 20.51 and 20.52 show how an intended eighth order Butterworth low pass can turn into a low pass filter with a Tschebyscheff characteristic, mainly due to capacitance changes from the partial filters.

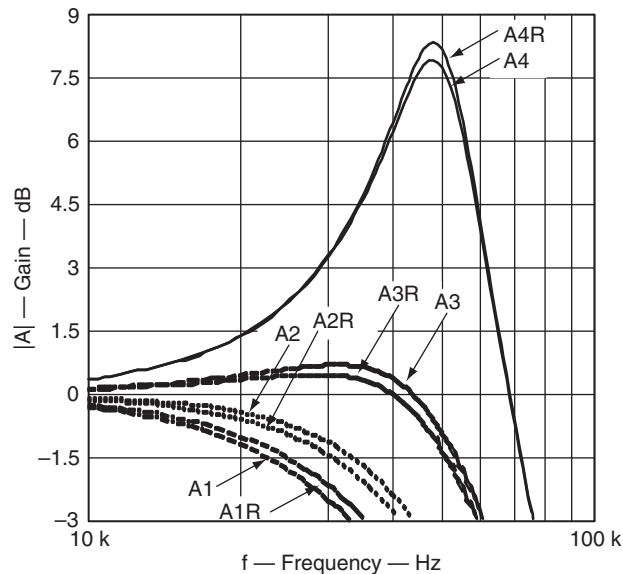


Figure 20.51: Differences in Q and f_c in the partial filters of an eighth order Butterworth low pass filter.

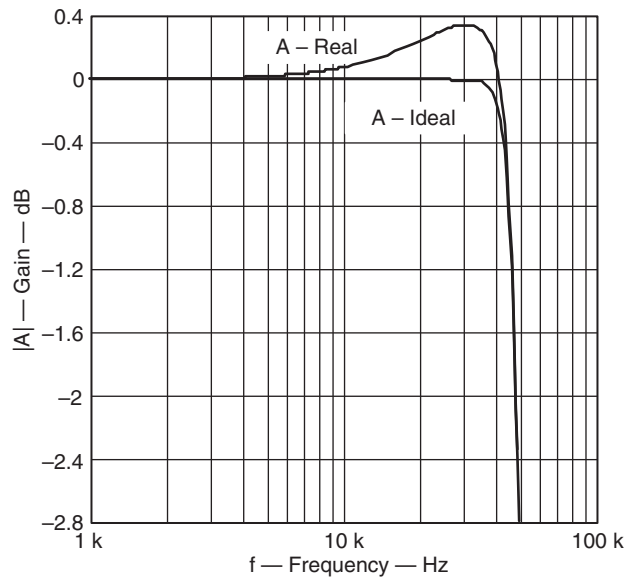


Figure 20.52: Modification of the intended Butterworth response to a Tschebyscheff type characteristic.

Figure 20.51 shows the differences between the ideal and the actual frequency responses of the four partial filters. The overall filter responses are shown in Figure 20.52. The difference between ideal and real response peaks with 0.35 dB at approximately 30 kHz, which is equivalent to an enormous 4.1% gain error, can be seen.

If this filter is intended for a data acquisition application, it could be used at best in a 4 bit system. In comparison, if the maximum full scale error of a 12 bit system is given with $\frac{1}{2}$ LSB, then maximum passband deviation would be -0.001 dB, or 0.012%.

To minimize the variations of f_C and Q , NPO (COG) ceramic capacitors are recommended for high performance filters. These capacitors hold their nominal value over a wide temperature and voltage range. The various temperature characteristics of ceramic capacitors are identified by a three symbol code, such as COG, X7R, Z5U, and Y5V.

COG type ceramic capacitors are the most precise. Their nominal values range from 0.5 pF to approximately 47 nF with initial tolerances from ± 0.25 pF for smaller values and up to $\pm 1\%$ for higher values. Their capacitance drift over temperature is typically 30 PPM/ $^{\circ}\text{C}$.

X7R type ceramic capacitors range from 100 pF to 2.2 μ F with an initial tolerance of +1% and a capacitance drift over temperature of $\pm 15\%$.

For higher values, tantalum electrolytic capacitors should be used.

Other precision capacitors are silver mica, metallized polycarbonate, and for high temperatures, polypropylene or polystyrene.

Since capacitor values are not as finely subdivided as resistor values, the capacitor values should be defined prior to selecting resistors. If precision capacitors are not available to provide an accurate filter response, then it is necessary to measure the individual capacitor values and calculate the resistors accordingly.

For high performance filters, 0.1% resistors are recommended.

20.8.3 Component Values

Resistor values should stay within the range of 1 k Ω to 100 k Ω . The lower limit avoids excessive current draw from the op amp output, which is particularly important for single supply op amps in power sensitive applications. Those amplifiers have typical output currents of between 1 mA and 5 mA. At a supply voltage of 5 V, this current translates to a minimum of 1 k Ω . The upper limit of 100 k Ω is to avoid excessive resistor noise.

Capacitor values can range from 1 nF to several microfarads. The lower limit avoids coming too close to parasitic capacitances. If the common mode input capacitance of the op amp, used in a Sallen-Key filter section, is close to 0.25% of C_1 , ($C_1/400$), it must be considered for accurate filter response. The MFB topology, in comparison, does not require input capacitance compensation.

20.8.4 Op Amp Selection

The most important op amp parameter for proper filter functionality is the unity gain bandwidth. In general, the open loop gain (A_{OL}) should be 100 times (40 dB above) the peak gain (Q) of a filter section to allow a maximum gain error of 1% (Figure 20.53).

The following equations are good rules of thumb to determine the necessary unity gain bandwidth of an op amp for an individual filter section.

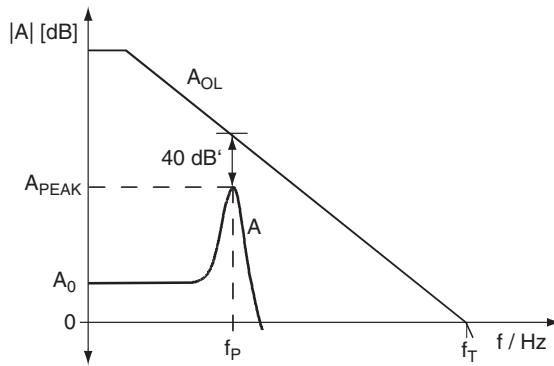


Figure 20.53: Open loop gain (A_{OL}) and filter response (A).

1. First order filter:

$$f_T = 100 \times \text{Gain} \times f_c$$

2. Second order filter ($Q < 1$):

$$f_T = 100 \times \text{Gain} \times f_c \times k_i \text{ with } k_i = f_{ci}/f_c$$

3. Second order filter ($Q > 1$):

$$f_T = 100 \times \text{Gain} \times \frac{f_c}{a_i} \sqrt{\frac{Q_i^2 - 0.5}{Q_i^2 - 0.25}}$$

For example, a fifth order, 10 kHz, Tschebyscheff low pass filter with 3 dB passband ripple and a DC gain of $A_0 = 2$ has its worst case Q in the third filter section. With $Q_3 = 8.82$ and $a_3 = 0.1172$, the op amp needs to have a unity gain bandwidth of

$$f_T = 100 \times 2 \times \frac{10 \text{ kHz}}{0.1172} \sqrt{\frac{8.82^2 - 0.5}{8.82^2 - 0.25}} = 17 \text{ MHz}$$

In comparison, a fifth order unity gain, 10 kHz, Butterworth low pass filter has a worst case Q of $Q_3 = 1.62$; $a_3 = 0.618$. Due to the lower Q value, f_T is also lower and calculates to only

$$f_T = 100 \times \frac{10 \text{ kHz}}{0.618} \sqrt{\frac{1.62^2 - 0.5}{1.62^2 - 0.25}} = 1.5 \text{ MHz}$$

Besides good DC performance, low noise, and low signal distortion, another important parameter that determines the speed of an op amp is the slew rate (SR). For adequate full power response, the slew rate must be greater than

$$\text{SR} = \pi \times V_{\text{PP}} \times fC$$

For example, a single supply, 100 kHz filter with 5 V_{PP} output requires a slew rate of at least

$$\text{SR} = \pi \times 5\text{V} \times 100 \text{ kHz} = 1.57 \text{ V}/\mu\text{s}$$

Texas Instruments offers a wide range of op amps for high performance filters in single supply applications. [Table 20.6](#) provides a selection of single supply amplifiers sorted in order of rising slew rate.

Table 20.6: Single Supply Op Amp Selection Guide ($T_A = 25^\circ\text{C}$, $V_{\text{CC}} = 5 \text{ V}$)

Op amp	BW (MHz)	FPR (kHz)	SR (V/ μs)	V_{IO} (mV)	Noise (nV/Hz)
TLV2721	0.51	11	0.18	0.6	20
TLC2201A	1.8	159	2.5	0.6	8
TLV2771A	4.8	572	9	1.9	21
TLC071	10	1000	16	1.5	7
TLE2141	5.9	2800	45	0.5	10.5
THS4001	270	127 MHz (1 V_{PP})	400	6	7.5

20.9 Filter Coefficient Tables

The following tables contain the coefficients for the three filter types: Bessel and Butterworth (Tables 20.7 and 20.8) and Tschebyscheff. The Tschebyscheff tables (Tables 20.9 through 20.12) are split into categories for the following passband ripples: 0.5 dB, 1 dB, 2 dB, and 3 dB.

The table headers consist of the following quantities:

n = the filter order.

i = the number of the partial filter.

a_i, b_i = are the filter coefficients.

k_i = the ratio of the corner frequency of a partial filter, f_{Ci} , to the corner frequency of the overall filter, f_C . This ratio is used to determine the unity gain bandwidth of the op amp as well as to simplify the test of a filter design by measuring f_{Ci} and comparing it to f_C .

Q_i = the quality factor of the partial filter.

f_i/f_C = the ratio is used for test purposes of the all pass filters, where f_i is the frequency, at which the phase is 180° for a second order filter, respectively, 90° for a first order all pass filter.

T_{gr0} = the normalized group delay of the overall all pass filter.

Table 20.7: Bessel Coefficients

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.3617	0.6180	1.000	0.58
3	1	0.7560	0.0000	1.323	—
	2	0.9996	0.4772	1.414	0.69
4	1	1.3397	0.4889	0.978	0.52
	2	0.7743	0.3890	1.797	0.81

Table 20.7: Bessel Coefficients (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
5	1	0.6656	0.0000	1.502	—
	2	1.1402	0.4128	1.184	0.56
	3	0.6216	0.3245	2.138	0.92
6	1	1.2217	0.3887	1.063	0.51
	2	0.9686	0.3505	1.431	0.61
	3	0.5131	0.2756	2.447	1.02
7	1	0.5937	0.0000	1.648	—
	2	1.0944	0.3395	1.207	0.53
	3	0.8304	0.3011	1.695	0.66
	4	0.4332	0.2381	2.731	1.13
8	1	1.1112	0.3162	1.164	0.51
	2	0.9754	0.2979	1.381	0.56
	3	0.7202	0.2621	1.963	0.71
	4	0.3728	0.2087	2.992	1.23
9	1	0.5386	0.0000	1.857	—
	2	1.0244	0.2834	1.277	0.52
	3	0.8710	0.2636	1.574	0.59
	4	0.6320	0.2311	2.226	0.76
	5	0.3257	0.1854	3.237	1.32
10	1	1.0215	0.2650	1.264	0.50
	2	0.9393	0.2549	1.412	0.54
	3	0.7815	0.2351	1.780	0.62
	4	0.5604	0.2059	2.479	0.81
	5	0.2883	0.1665	3.466	1.42

Table 20.8: Butterworth Coefficients

n	i	a_i	b_i	$k_i = f_{Ci}/f_c$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.4142	1.0000	1.000	0.71
3	1	1.0000	0.0000	1.000	—
	2	1.0000	1.0000	1.272	1.00
4	1	1.8478	1.0000	0.719	0.54
	2	0.7654	1.0000	1.390	1.31
5	1	1.0000	0.0000	1.000	—
	2	1.6180	1.0000	0.859	0.62
	3	0.6180	1.0000	1.448	1.62
6	1	1.9319	1.0000	0.676	0.52
	2	1.4142	1.0000	1.000	0.71
	3	0.5176	1.0000	1.479	1.93
7	1	1.0000	0.0000	1.000	—
	2	1.8019	1.0000	0.745	0.55
	3	1.2470	1.0000	1.117	0.80
	4	0.4450	1.0000	1.499	2.25
8	1	1.9616	1.0000	0.661	0.51
	2	1.6629	1.0000	0.829	0.60
	3	1.1111	1.0000	1.206	0.90
	4	0.3902	1.0000	1.512	2.56
9	1	1.0000	0.0000	1.000	—
	2	1.8794	1.0000	0.703	0.53
	3	1.5321	1.0000	0.917	0.65
	4	1.0000	1.0000	1.272	1.00
	5	0.3473	1.0000	1.521	2.88

Table 20.8: Butterworth Coefficients (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
10	1	1.9754	1.0000	0.655	0.51
	2	1.7820	1.0000	0.756	0.56
	3	1.4142	1.0000	1.000	0.71
	4	0.9080	1.0000	1.322	1.10
	5	0.3129	1.0000	1.527	3.20

Table 20.9: Tschhebyscheff Coefficients for 0.5 dB Passband Ripple

n	i	a_i	b_i	$k_i = f_{Gi}/f_C$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.3614	1.3827	1.000	0.86
3	1	1.8636	0.0000	0.537	—
	2	0.0640	1.1931	1.335	1.71
4	1	2.6282	3.4341	0.538	0.71
	2	0.3648	1.1509	1.419	2.94
5	1	2.9235	0.0000	0.342	—
	2	1.3025	2.3534	0.881	1.18
	3	0.2290	1.0833	1.480	4.54
6	1	3.8645	6.9797	0.366	0.68
	2	0.7528	1.8573	1.078	1.81
	3	0.1589	1.0711	1.495	6.51
7	1	4.0211	0.0000	0.249	—
	2	1.8729	4.1795	0.645	1.09
	3	0.4861	1.5676	1.208	2.58
	4	0.1156	1.0443	1.517	8.84

(Continued)

Table 20.9: Tschhebyscheff Coefficients for 0.5 dB Passband Ripple (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci} / f_C$	Q_i
8	1	5.1117	11.9607	0.276	0.68
	2	1.0639	2.9365	0.844	1.61
	3	0.3439	1.4206	1.284	3.47
	4	0.0885	1.0407	1.521	11.53
9	1	5.1318	0.0000	0.195	—
	2	2.4283	6.6307	0.506	1.06
	3	0.6839	2.2908	0.989	2.21
	4	0.2559	1.3133	1.344	4.48
	5	0.0695	1.0272	1.532	14.58
10	1	6.3648	18.3695	0.222	0.67
	2	1.3582	4.3453	0.689	1.53
	3	0.4822	1.9440	1.091	2.89
	4	0.1994	1.2520	1.381	5.61
	5	0.0563	1.0263	1.533	17.99

Table 20.10: Tschhebyscheff Coefficients for 1 dB Passband Ripple

n	i	a_i	b_i	$k_i = f_{Ci} / f_C$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.3022	1.5515	1.000	0.96
3	1	2.2156	0.0000	0.451	—
	2	0.5442	1.2057	1.353	2.02
4	1	2.5904	4.1301	0.540	0.78
	2	0.3039	1.1697	1.417	3.56

Table 20.10: Tschebyscheff Coefficients for 1 dB Passband Ripple (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci}/f_c$	Q_i
5	1	3.5711	0.0000	0.280	—
	2	1.1280	2.4896	0.894	1.40
	3	0.1872	1.0814	1.486	5.56
6	1	3.8437	8.5529	0.366	0.76
	2	0.6292	1.9124	1.082	2.20
	3	0.1296	1.0766	1.493	8.00
7	1	4.9520	0.0000	0.202	—
	2	1.6338	4.4899	0.655	1.30
	3	0.3987	1.5834	1.213	3.16
	4	0.0937	1.0432	1.520	10.90
8	1	5.1019	14.7608	0.276	0.75
	2	0.8916	3.0426	0.849	1.96
	3	0.2806	1.4334	1.285	4.27
	4	0.0717	1.0432	1.520	14.24
9	1	6.3415	0.0000	0.158	—
	2	2.1252	7.1711	0.514	1.26
	3	0.5624	2.3278	0.994	2.71
	4	0.2076	1.3166	1.346	5.53
	5	0.0562	1.0258	1.533	18.03
10	1	6.3634	22.7468	0.221	0.75
	2	1.1399	4.5167	0.694	1.86
	3	0.3939	1.9665	1.093	3.56
	4	0.1616	1.2569	1.381	6.94
	5	0.0455	1.0277	1.532	22.26

Table 20.11: Tschebyscheff Coefficients for 2 dB Passband Ripple

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.1813	1.7775	1.000	1.13
3	1	2.7994	0.0000	0.357	—
	2	0.4300	1.2036	1.378	2.55
4	1	2.4025	4.9862	0.550	0.93
	2	0.2374	1.1896	1.413	4.59
5	1	4.6345	0.0000	0.216	—
	2	0.9090	2.6036	0.908	1.78
	3	0.1434	1.0750	1.493	7.23
6	1	3.5880	10.4648	0.373	0.90
	2	0.4925	1.9622	1.085	2.84
	3	0.0995	1.0826	1.491	10.46
7	1	6.4760	0.0000	0.154	—
	2	1.3258	4.7649	0.665	1.65
	3	0.3067	1.5927	1.218	4.12
8	1	4.7743	18.1510	0.282	0.89
	2	0.6991	3.1353	0.853	2.53
	3	0.2153	1.4449	1.285	5.58
	4	0.0547	1.0461	1.518	18.39
9	1	8.3198	0.0000	0.120	—
	2	1.7299	7.6580	0.522	1.60
	3	0.4337	2.3549	0.998	3.54
	4	0.1583	1.3174	1.349	7.25
	5	0.0427	1.0232	1.536	23.68

Table 20.11: Tschebyscheff Coefficients for 2 dB Passband Ripple (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
10	1	5.9618	28.0376	0.226	0.89
	2	0.8947	4.6644	0.697	2.41
	3	0.3023	1.9858	1.094	4.66
	4	0.1233	1.2614	1.380	9.11
	5	0.0347	1.0294	1.531	29.27

Table 20.12: Tschebyscheff Coefficients for 3 dB Passband Ripple

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
1	1	1.0000	0.0000	1.000	—
2	1	1.0650	1.9305	1.000	1.30
3	1	3.3496	0.0000	0.299	—
	2	0.3559	1.1923	1.396	3.07
4	1	2.1853	5.5339	0.557	1.08
	2	0.1964	1.2009	1.410	5.58
5	1	5.6334	0.0000	0.178	—
	2	0.7620	2.6530	0.917	2.14
	3	0.1172	1.0686	1.500	8.82
6	1	3.2721	11.6773	0.379	1.04
	2	0.4077	1.9873	1.086	3.46
	3	0.0815	1.0861	1.489	12.78
7	1	7.9064	0.0000	0.126	—
	2	1.1159	4.8963	0.670	1.98
	3	0.2515	1.5944	1.222	5.02
	4	0.0582	1.0348	1.527	17.46

(Continued)

Table 20.12: Tschebyscheff Coefficients for 3 dB Passband Ripple (Cont'd)

n	i	a_i	b_i	$k_i = f_{Ci}/f_C$	Q_i
8	1	4.3583	20.2948	0.286	1.03
	2	0.5791	3.1808	0.855	3.08
	3	0.1765	1.4507	1.285	6.83
	4	0.0448	1.0478	1.517	22.87
9	1	10.1759	0.0000	0.098	—
	2	1.4585	7.8971	0.526	1.93
	3	0.3561	2.3651	1.001	4.32
	4	0.1294	1.3165	1.351	8.87
	5	0.0348	1.0210	1.537	29.00
10	1	5.4449	31.3788	0.230	1.03
	2	0.7414	4.7363	0.699	2.94
	3	0.2479	1.9952	1.094	5.70
	4	0.1008	1.2638	1.380	11.15
	5	0.0283	1.0304	1.530	35.85

Table 20.13: All Pass Coefficients

n	i	a_i	b_i	f_i/f_C	Q_i	T_{gr0}
1	1	0.6436	0.0000	1.554	—	0.2049
2	1	1.6278	0.8832	1.064	0.58	0.5181
3	1	1.1415	0.0000	0.876	—	0.8437
	2	1.5092	1.0877	0.959	0.69	
4	1	2.3370	1.4878	0.820	0.52	1.1738
	2	1.3506	1.1837	0.919	0.81	

Table 20.13: All Pass Coefficients (Cont'd)

n	i	a_i	b_i	f_i/f_c	Q_i	T_{gr0}
5	1	1.2974	0.0000	0.771	—	1.5060
	2	2.2224	1.5685	0.798	0.56	
	3	1.2116	1.2330	0.901	0.92	
6	1	2.6117	1.7763	0.750	0.51	1.8395
	2	2.0706	1.6015	0.790	0.61	
	3	1.0967	1.2596	0.891	1.02	
7	1	1.3735	0.0000	0.728	—	2.1737
	2	2.5320	1.8169	0.742	0.53	
	3	1.9211	1.6116	0.788	0.66	
	4	1.0023	1.2743	0.886	1.13	
8	1	2.7541	1.9420	0.718	0.51	2.5084
	2	2.4174	1.8300	0.739	0.56	
	3	1.7850	1.6101	0.788	0.71	
	4	0.9239	1.2822	0.883	1.23	
9	1	1.4186	0.0000	0.705	—	2.8434
	2	2.6979	1.9659	0.713	0.52	
	3	2.2940	1.8282	0.740	0.59	
	4	1.6644	1.6027	0.790	0.76	
	5	0.8579	1.2862	0.882	1.32	
10	1	2.8406	2.0490	0.699	0.50	3.1786
	2	2.6120	1.9714	0.712	0.54	
	3	2.1733	1.8184	0.742	0.62	
	4	1.5583	1.5923	0.792	0.81	
	5	0.8018	1.2877	0.881	1.42	

References

1. Johnson, D., and J. Hilburn. (1975). *Rapid Practical Designs of Active Filters*, New York: Wiley.
2. Tietze, U., and C. Schenk. (1980). *Halbleiterschaltungstechnik*. Berlin: Springer-Verlag.
3. Berlin, H. (1979). *Design of Active Filters with Experiments*. Howard W. Sams.
4. Van Falkenburg, M. (1982). *Analog Filter Design*. Oxford University Press.
5. Franko, S. (1988). *Design with Operational Amplifiers and Analog Integrated Circuits*. New York: McGraw-Hill.